AmpScape: a benchmark for learned surrogates of circuit-theoretic landscape connectivity

Anonymous Author(s)
Affiliation
Address
email

Abstract

Circuit-theoretic connectivity solves a high-contrast elliptic operator with per-sample point sources on rasters up to 2048^2; it is used in hundreds of conservation studies, and its omnidirectional variant (Omniscape) and many-query uses—sensitivity analysis, scenario comparison, resistance-surface optimisation—make the solver the bottleneck. We introduce AmpScape, a benchmark for learned surrogates of this operator: 174,400 resistance landscapes on five tiers from 128^2 to 2048^2 pixels (60 % synthetic, 40 % real tiles built from open global layers), each paired with exact Circuitscape.jl and Omniscape.jl outputs: 920,372 configuration rows and 1,157 GB, every pairwise row with its solver residual and the Omniscape approximation measured against exact references. The benchmark defines pairwise, advanced-mode and omnidirectional tasks with pixel, ranking, non-source and physics metrics, spatial and distributional out-of-distribution splits and a zero-shot scale-transfer split. Baselines from five model families (three seeds, tuned per tier on validation loss alone) show a speed–accuracy–robustness trade-off: the best surrogate reaches rel-L2 0.041 / 0.042 / 0.081 on the omnidirectional task at 10^4–$10^5\times$ the speed of the single-core solver (22,916$\times$ against the solver on 24 threads at L, break-even after 37 maps including training), pairwise current is much harder (0.114 / 0.187 / 0.299) and there a converged preconditioned-CG solve of every pair costs 0.6–17 s on one core and two iterations already beat every learned model on the median landscape, zero-shot transfer to larger tiers fails on magnitude by exactly the factor the window geometry predicts, and only rankings survive. A many-query demonstration reproduces every study-level conclusion of a 160-map screening study at 11,077$\times$ lower cost than a single-core solver run; its persistent pinch points (precision 0.657 under bf16 inference, 0.791 in fp32) still need solver verification. Data, code, harness, leaderboard and Croissant metadata are released under CC BY 4.0 and MIT.

1 Introduction

Circuit-theoretic connectivity [32, 33] turns a resistance raster into a resistor network and reads movement potential from the current that flows between habitat patches; the method paper has 1,936 Scopus citations and a review counted 459 applications in four years [10]. For machine learning it is a large, structured instance of elliptic operator learning: $\nabla \cdot (\sigma \nabla \phi) = f$ with coefficient contrast up to 10^6, singular sources, and decisions that depend on rankings rather than magnitudes. Two things make its solvers expensive. The omnidirectional variant, Omniscape [35, 25], solves one circuit per source block inside a moving window around every pixel: documented production runs took 140 h on 55 CPUs per species [9] or 196 h for one regional map [19], and in our own reference generation the median omnidirectional solve rises from 52 s per landscape at 128^2 to 9,821 s at 2048^2. A single pairwise solve is cheap (a median of 0 to 63 s across our tiers), but resistance surfaces are

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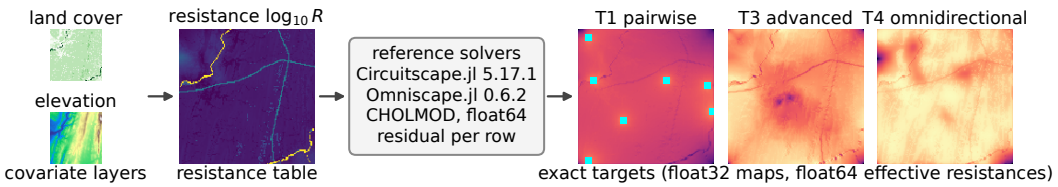


Figure 1: AmpScape in one view. Open covariate layers (or a synthetic generator) and a resistance table give a resistance raster; the reference solvers produce exact pairwise, advanced-mode and omnidirectional current maps, which the tasks turn into learning targets. Five tiers, the task menu, the split design and the metric families are listed below the pipeline.

uncertain inputs [52], so the solver is called repeatedly inside optimisation loops, scenario sweeps and uncertainty analyses [42, 17, 6]: the *many-query* pattern is the cost. Public operator-learning benchmarks have contrasts of 6, uniform forcing and grids far below 2048^2 (Section 2).

Contributions. (i) *The dataset*: 174,400 landscapes on five tiers from 128^2 at 100 m to 2048^2 at 1 km, 60 % synthetic with a hard-case stratum and 40 % real tiles from open global layers, each solved under five resistance tables (Figure 1, Table 1). (ii) *A reference-solver protocol with measured fidelity*: every pairwise and advanced-mode row carries its Kirchhoff residual, and the block-centred Omniscape targets are measured against exact block-1 references (mean rel-L2 0.029 at M, 0.030 at L) that become the official evaluation surface at those tiers. (iii) *A harness* with pixel, ranking, non-source-pixel, physics and efficiency metrics, a framework-agnostic prediction format and a leaderboard. (iv) *Baselines*: five model families, three seeds, configurations tuned per tier and selected per task on validation loss alone, with scale-transfer, OOD, data-scaling and solver-acceleration studies. (v) *A many-query demonstration* on a 160-map screening study.

The headline is a trade-off, not a replacement. On the omnidirectional task the best surrogate reaches rel-L2 0.041, 0.042 and 0.081 at S, M and L at median speed-ups of 167,020×, 32,789× and 111,998× over the production solver on one core (14,235× and 22,916× against the 24-thread solver at M and L, Appendix C). Pairwise current is harder for every model (best rel-L2 0.114 / 0.187 / 0.299), zero-shot transfer to XL and XXL fails on magnitude by exactly the factor the larger window injects (rel-L2 ≈ 0.50 and ≈ 0.75, Appendix A), and ranking metrics survive where magnitudes do not. AmpScape thus measures *when* a learned surrogate of a high-contrast elliptic operator is the right tool, and when a solver remains preferable.

2 Related work

Circuit theory as ground truth, and its cost. McRae's isolation by resistance [32] and its connectivity tool [33] solve Kirchhoff's equations on the raster graph; Circuitscape.jl uses CG+AMG and CHOLMOD solvers [1, 14], Omniscape applies it in a moving window around every source pixel [35, 25], gflow parallelises pairwise solves [26], ConScape covers the randomised-shortest-path continuum [50, 41], and current density has been computed at global extent [7]. Nearly-linear-time solvers and effective-resistance sketches exist in theory [23, 46]; production uses CG+AMG or CHOLMOD, called repeatedly because resistance surfaces are uncertain [52, 42, 17, 6]. Practitioners report 140 h on 55 CPUs per species [9] and 196 h for one regional map [19]; the standard mitigation, `block_size`, is documented as giving "only negligable [sic] differences" [40] and set to 10 % of the window radius in practice [3]; AmpScape measures it against exact block-1 solutions (Section 4).

Where to intervene, and ML for connectivity. Barrier Mapper [34], Pinchpoint Mapper [29], Zonation [36, 37], Marxan Connect [8] and GECOT [16] prioritise interventions; analytical sensitivities [49] and differentiable, GPU-accelerated distances [5] show that exact solvers are differentiable and accelerator-capable, so the case for a surrogate is amortised cost across many queries. Machine learning has been used *around* connectivity [11, 45], but our searches found no learned surrogate of Circuitscape or Omniscape and no public benchmark of solver-computed circuit-theory outputs.

Learned surrogates for elliptic problems; benchmark design. Circuitscape's system discretises $\nabla \cdot (\sigma \nabla \phi) = f$ with point sources and grounds, so the closest lineage is operator learning on Darcy flow—FNO [28], PDEBench [47], the neuraloperator library [22], U-Net baselines [44, 13], graph networks [2, 43]—with multiscale, high-contrast coefficients as the recognised hard case [30, 18, 51, 15, 48] and learned preconditioners that accelerate rather than replace the solver [24, 27]. The delta is in the data: Darcy-rough has "a contrast of 6" with "$f(x) \equiv 1$" [51]; AmpScape has real coefficient fields, contrast to 10^6, singular sources, grids to 2048^2 and geographic OOD splits. The release follows PGLearn and OPFData [21, 31, 20] (reference solver, instance families, complete solutions), PDEBench, The Well and AirfRANS [39, 4] (HDF5 shards, cards, OOD tasks) and Datasheets and the track call [12, 38].

3 Problem and tasks

Each sample pairs a resistance raster R ($H \times W$, $R \geq 1$, NoData masked) and a source configuration with the exact output of Circuitscape.jl 5.17.1 or Omniscape.jl 0.6.2: one node per valid pixel, 8-neighbour edges of conductance $g_{ij} = (g_i + g_j)/2$ with $g = 1/R$, voltages v solving $Lv = b$ for the injection vector b, and Circuitscape's node current density as the stored map. The reference solver is CHOLMOD in double precision, every sample records its residual $\|Lv - b\|_2 / \|b\|_2$, and outputs are stored raw (float32 maps, float64 effective resistances).

T1, pairwise current: Circuitscape `pairwise` mode with $K \in [2, 8]$ single-pixel focal nodes and all $K(K-1)/2$ pairs (node i grounded, 1 A injected at node j); target `cum_current` ($H \times W$), with the $K \times K$ effective-resistance matrix as the **T2** target. **T1W** uses $K = 2$ edge strips as focal regions (strip pixels excluded from the metrics), **T1R** habitat patches as short-circuited regions. T2, T1W and T1R are released with data, solver references and harness support but *without learned baselines*; Section 7 trains on T1, T3 and T4. **T3**, advanced mode: source raster values are injected currents and grounds are ideal; targets `current` and `voltage`. **T4**, omnidirectional connectivity: Omniscape grounds in turn every pixel with source strength above `source_threshold` (= 0) on a grid of stride `block_size`, injects current $\propto S$ from all sources within `radius` px and sums the window maps (sources: inverse-resistance suitability above the 0.5 quantile, rescaled to a maximum of 1). Production radius / block per tier: S 16 / 1, M 32 / 3, L 64 / 5, XL 128 / 11, XXL 256 / 25 px (block = largest odd integer $\leq$ radius/10), `correct_artifacts = true`; block 1 is exact, larger blocks give the block-centred map, a measured approximation (Section 4).

Inputs and target for learning. Baselines receive standardised log-resistance (train-only statistics, 0 at NoData), the NoData mask and the task's source channels, and predict $\log_{10}(C + \epsilon \cdot \max C)$, $\epsilon = 10^{-6}$, under a masked MSE: scale-invariant, 6 decades of range below the map maximum.

4 The dataset

Landscapes. A landscape is one resistance raster; each of its source configurations is a configuration row (Table 1). *Real landscapes* are covariate stacks from ESA WorldCover, Copernicus DEM, GRIP4 roads, HydroRIVERS and gHM (13 958 tiles, 69 790 resistance rasters); each tile yields five landscapes, one per resistance table—four expert-style tables and one seeded random table per tile, with the `forest_bird` table ($r_{\max}$ 100 versus 1000) held out. Class ordering and term structure follow the literature; the numeric values are AmpScape's own, not species-calibrated truth. *Synthetic landscapes* come from neutral landscape models and random fields, with 20 % in a hard stratum (contrast 10^5 and 10^6, $r_{\max}$-saturated, narrow corridors, large NoData); contrast 10^6 is reserved for the OOD contrast set (Figure 2).

Task groups and layout. Every landscape receives `points` (T1), two wall-to-wall strips (T1W), an `advanced` configuration (T3) and an Omniscape configuration (T4); landscapes with ≥ 2 eligible

Table 1: AmpScape v1.0 per tier. A landscape is one resistance raster; each of its source configurations is a configuration row. The 25 rows failing quality control (24 landscapes, all synthetic at contrast $\geq 10^4$) stay in the index with their flags and are excluded from the split lists.

tier	raster · pixel	landscapes	configuration rows	synthetic / real	GB on the Hub	rows failing QC
S	128^2 · 100 m	100,000	527,281	60,000 / 40,000	142.4	1
M	256^2 · 100 m	50,000	264,128	30,000 / 20,000	251.7	4
L	512^2 · 200 m	20,000	105,949	12,000 / 8,000	380.9	11
XL	1024^2 · 500 m	4,000	21,006	2,400 / 1,600	281.7	7
XXL	2048^2 · 1 km	400	2,008	210 / 190	100.7	2
total		174,400	920,372		1157.4	25

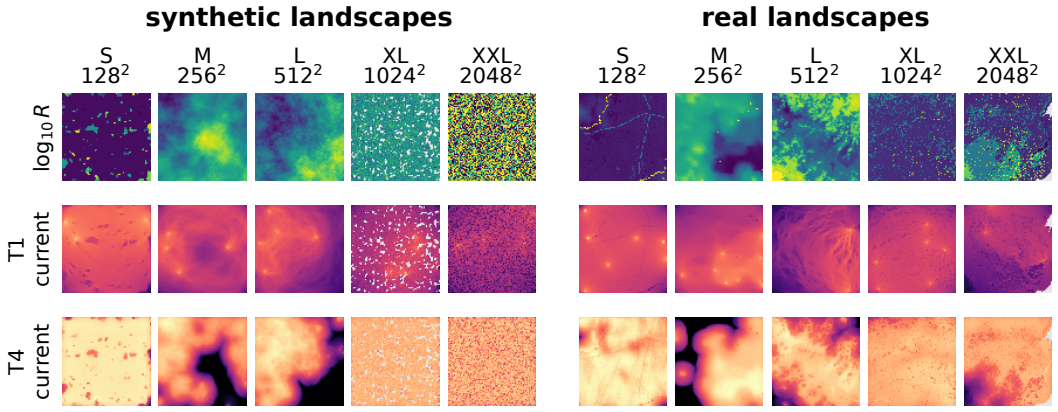


Figure 2: Data gallery: one synthetic (left) and one real (right) landscape per tier, columns S (128^2) to XXL (2048^2); rows are $\log_{10}$ resistance, the T1 pairwise current map and the T4 omnidirectional current map ($\log_{10}$ scale, NoData grey). Every sample is from `test_id` except the S real tile (`mini` subset, training split).

habitat patches also get `regions` (T1R). Shards are per tier and task group with a per-tier index, split lists and `croissant.json`, so any tier or task group downloads alone; four nested subsets—`mini` (0.6 GB), `lite` ($\approx$ 26 GB), `core` (115.5 GB), `full` (1 157.4 GB)—each have an index column, split lists and a Croissant FileSet. Hosting: `Xirro/AmpScape`, CC BY 4.0, Croissant 1.0 with RAI fields.

Reference solver, precision pass, Omniscape fidelity. All T1/T1W/T1R/T3 targets are computed with the direct CHOLMOD solver in double precision, with the Kirchhoff residual recorded from full-precision voltages (QC threshold 10^{-6}). A post-run precision pass re-solved 129,722 rows (residual above 10^{-9}, unmeasured, CG+AMG-solved or QC-failed) with CHOLMOD plus refinement: 3,861 remain between 10^{-9} and 10^{-6} and 20 are flagged and excluded with the other QC failures (25 rows, 24 landscapes; Appendix G). T4 targets use Omniscape.jl 0.6.2 with artefact correction. Because `block_size` is part of the method, it is fixed per tier by the rule above and the block-centred approximation is *measured*: on the 1 000-sample M reference the production block-3 target deviates from the exact block-1 map by mean rel-L2 0.029 with 4.7 % of samples above 5 %; the L target (block 5, 120 samples) by mean 0.030 (max 0.203, at contrast 10^6); block 1 costs a median 7.2× (M) and 19.8× (L) the production solve per landscape. These block-1 references are the official T4 evaluation surface at M and L (Appendix F). Every tier was audited on the Hub, and regeneration is bitwise reproducible on one CPU model ($\leq 1.2 \times 10^{-9}$ relative across models).

5 Splits and out-of-distribution design

Splits are assigned by the *unit of leakage*—the tile for real landscapes, the seed family for synthetic ones—at base fractions 0.8 / 0.1 / 0.1, with hold-outs applied before the draw. Because an S tile physically sits inside XL/XXL footprints, the globe is partitioned once into 104 equal-width macrocells with one seeded assignment applied at every tier (70 land cells: 56 train / 7 val / 7 test_id);

no train/val tile box intersects any test tile box at any tier, XXL stays test-only, seed ranges are tier-disjoint, and 25 % of XL landscapes are train/val (counts in Appendix I).

OOD sets (one boolean index column each): `region`—real tiles in the Montane Grasslands & Shrublands or Mangroves biomes or the Australasia realm; `table`—every landscape built with the `forest_bird` table; `contrast`—synthetic contrast 10^6; `scale`—XL/XXL evaluated by models trained at $\leq$ L, with a `strict` subset of 6 XXL real tiles that overlap no training tile at any tier; `synth2real`—a loader flag restricting training to synthetic landscapes; `published`—45 S tiles and 1 XXL tile with published resistance surfaces. Held-out-table and held-out-contrast landscapes that the base draw would place in train/val are demoted to `test_ood`.

6 Metrics and evaluation protocol

Maps are compared in $\log_{10}(C + \epsilon \cdot \max C)$. *Pixel*: relative L2 on raw currents (rel-L2, the primary metric and the one every headline and table ranks by), MAE in $\log_{10}$-ϵ space (the training-loss space), MSE, SSIM/PSNR; NoData and T1W strip pixels masked. *Domain*: top-q % IoU of the q % highest-current valid pixels ($q \in \{1, 5, 10\}$); corridor Dice above the 10 % quantile; pinch-point recall (target pinch points = local maxima in a 7×7 window within the top 5 %, hit = a predicted local maximum within 3 px); Spearman rank correlation—conventions, not validated ecological criteria. *Non-source pixels*: currents are singular at injection pixels, so rel-L2, MAE and top-q IoU are also computed outside source, focal and ground pixels plus a 2-px halo. *Physics*: Kirchhoff residual of a predicted voltage, focal-current error, non-negativity, throughput error. *Efficiency*: speed-up = solver wall time / inference time, batch 1 at XL/XXL; solver wall times are the production setting, one solve per core with Omniscape's own parallelisation off (multi-thread timings and the training cost in Appendix C). *Acceleration*: AMG-preconditioned CG iterations and wall time to the reference tolerance from a predicted voltage versus from zero.

Results are reported on `test_id` and every OOD split, three seeds per configuration (mean $\pm$ sample std), with early stopping on `val`. rel-L2 is additionally reported as the median over seeds: it is a ratio of squared sums dominated by the few highest-current landscapes, so one seed that misses them moves the mean by more than the seed spread of every other metric (at L the three GNN seeds give 0.789, 0.925 and 1.108 while their MAE and Spearman agree within 0.02). Landscape-resampling 95 % intervals are within ± 0.016 of every Table 2 mean except the GNN on T1 at L (± 0.042), and every best-versus-runner-up order holds in all 2,000 resamples (Table 10). Predictions are framework-agnostic (Appendix J) and scored by one harness; at M and L the primary T4 metrics are computed against the exact block-1 map on the reference subsets (400 / 300 / 300 landscapes per split at M; 48 / 32 / 40 at L). The Hub leaderboard accepts the same format.

7 Baselines and results

Models, protocol, configuration selection. All learned baselines share the inputs and target of Section 3. **U-Net**: 4-level encoder–decoder, width 32 (7.8 M parameters), tuned width-64 variant `w64` (31.0 M). **FNO**: 64 retained modes plus a distance-to-source channel, width 32 (33.6 M), tuned width-48 variant `w48` (75.5 M). **ViT**: patch-4 embedding, global attention (dim 192, depth 6), a learned 32×32 positional embedding interpolated to other grids (3.0 M); tuned `p16g16` variant with patch 16 at M; at L both larger-patch candidates were worse on validation (47.8 % and 63.5 %), so the resolution-inappropriate configuration stays official, a negative result. **GNN**: grid-as-graph message passing with a $4\times$-coarsened second level (`ms`, 0.2 M), tuned $8\times$-coarsened variant `ms8`. **MgNO** [18]: the authors' MIT release (0.58 M), three seeds at S and M on T1 and T4. Non-learned rows: Omniscape *block-size rows* for T4 and, for T1 and T3, a coarsen-$\times 4$-and-upsample solve and the reference solver's preconditioned CG stopped after k iterations (Appendix C). Each run trains one model on one task at one tier for 30 epochs with AdamW, a cosine schedule, patience 8 and a fixed seed, three seeds per (model, task, tier), with train-only normalisation statistics of the training tier at every evaluation tier. One alternative configuration per model was trained at M and L on T1 (seed 1) and compared on validation loss; the winners were re-trained with three seeds on every task, and the official configuration is chosen *per model* $\times$ *task* $\times$ *tier on validation loss alone*—tuned only if its mean best validation loss over seeds beats the pre-tuning mean by more than the pre-tuning standard error (std/$\sqrt{3}$); test metrics are never read and no decision is carried across tasks. Twelve of the fifteen decisions went to the tuned configuration (Appendix B).

Table 2: Relative L2 on `test_id` at the training tier, mean ± std over three seeds with the median over seeds in brackets (official configuration per model × task × tier; MgNO: three seeds at S and M, not ranked because 4 of its 6 T4 runs diverged, see text). Best three-seed model per task and tier in bold; lower is better. The complete metric set (MAE, top-5 % IoU, pinch recall, Spearman, speed-up, training cost) is Table 3.

task	model	S	M	L
T1	U-Net	**0.114 ± 0.007 [0.111]**	**0.187 ± 0.022 [0.187]**	0.397 ± 0.011 [0.392]
T1	FNO	0.199 ± 0.002 [0.199]	0.208 ± 0.003 [0.209]	**0.299 ± 0.011 [0.297]**
T1	ViT	0.218 ± 0.028 [0.229]	0.306 ± 0.013 [0.299]	0.648 ± 0.033 [0.644]
T1	GNN	0.561 ± 0.005 [0.563]	0.607 ± 0.027 [0.600]	0.941 ± 0.160 [0.925]
T1	MgNO	0.444 ± 0.238 [0.419]	0.376 ± 0.044 [0.389]	–
T3	U-Net	**0.111 ± 0.005 [0.110]**	**0.202 ± 0.005 [0.202]**	0.379 ± 0.012 [0.374]
T3	FNO	0.252 ± 0.001 [0.252]	0.255 ± 0.003 [0.254]	**0.323 ± 0.002 [0.323]**
T4	U-Net	**0.041 ± 0.001 [0.041]**	**0.042 ± 0.001 [0.041]**	**0.081 ± 0.001 [0.081]**
T4	FNO	0.051 ± 0.001 [0.051]	0.073 ± 0.000 [0.073]	0.116 ± 0.000 [0.116]
T4	ViT	0.052 ± 0.001 [0.052]	0.094 ± 0.004 [0.093]	0.159 ± 0.004 [0.161]
T4	GNN	0.066 ± 0.000 [0.066]	0.099 ± 0.001 [0.099]	0.142 ± 0.001 [0.142]
T4	MgNO	0.358 ± 0.544 [0.045]	0.184 ± 0.021 [0.189]	–

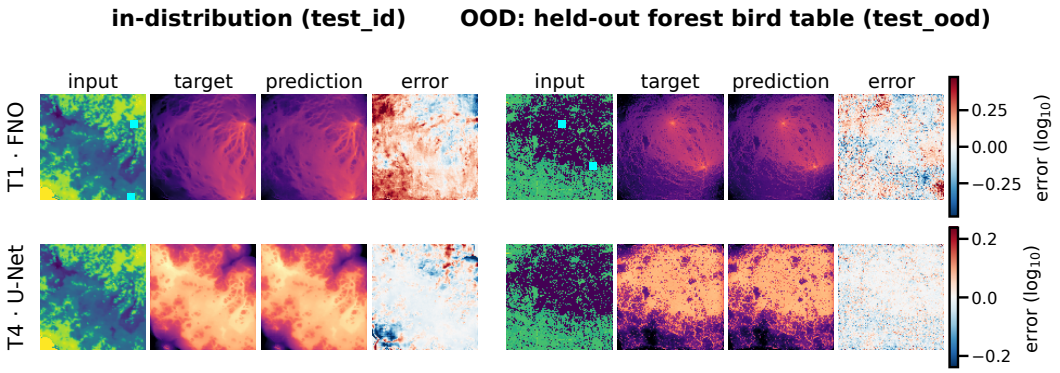


Figure 3: Qualitative results at L for the official T1 model (FNO w48, top) and T4 model (U-Net, bottom) on a real `test_id` landscape (left) and a `test_ood` landscape built with the held-out `forest_bird` table (right): input $\log_{10} R$ with the T1 focal nodes in cyan, exact target, prediction and the error in $\log_{10}$ units (one colour scale per row). Pairwise errors concentrate along the long-range flow paths; omnidirectional errors are local and small except at the NoData boundary.

In-distribution results. Table 2, Figure 4 and, qualitatively, Figure 3 give rel-L2 on `test_id`. First, the task families differ: T4 is the easier learning problem and the one with the real compute problem—the U-Net stays at rel-L2 0.041–0.081 from S to L with Spearman 0.996 / 0.996 / 0.992 at median speed-ups of 167,020× / 32,789× / 111,998× over the production Omniscape run on one core (training paid back after 152 and 37 maps at M and L against the 24-thread solver, Appendix C)—whereas T1 and T3, globally coupled through one sparse solve, are harder for every model and give speed-ups of only 10^2–10^3 (U-Net T1 625× / 537× / 691×) because a full-raster pairwise solve is already cheap. Second, error grows with tier for every model and task (Figure 4). Third, ranking metrics degrade more slowly than magnitudes (T4 U-Net, S to L: rel-L2 0.041 → 0.081, pinch recall 0.918 → 0.825, Spearman 0.996 → 0.992).

The families fail in different places. The U-Net is strongest on T1 and T4 at S and M but loses its T1 lead to the FNO at L (0.397 vs 0.299); the FNO has the best T1 result at L and the best transfer; the ViT is competitive at S (T1 0.218, T4 0.052) and resolution-inappropriate at L even after tuning (T1 0.648, top-5 % IoU 0.146); the GNN is far behind on T1 (0.561 / 0.607 / 0.941) but competitive on the locally windowed T4 (0.066 / 0.099 / 0.142) at 5.06–9.44 GPU-h per run against 0.79–1.97 for the U-Net; under the shared protocol (AdamW 10^{-3}, bf16 autocast) MgNO's training loss explodes within four epochs in 4 of 6 T4 runs and 2 of 6 T1 runs, so its seed means carry that instability (T4 S 0.358, median 0.045 against the U-Net's 0.041). No family wins on accuracy, scale robustness, OOD robustness and cost at once; that trade-off is the result.

T4 against the exact reference. For the reference subsets we computed exact block-1 maps, so solver approximations and learned models share one error-versus-cost axis (Figure 5; rows in Appendix B). The production targets approximate the exact map well (rel-L2 0.0291 at M, 0.0321 at L); Omniscape's artefact correction matters more than the block size (block 3 without correction, 0.1105, is worse than block 7 with it, 0.0983, at M); and the learned models occupy a different part of the frontier rather than dominating it: at M the U-Net's 0.0510 lies between the production solver (0.0291 at 111 s) and block 7 (0.0983 at 25.5 s) at 0.00328 s per landscape, at L its 0.0850 matches block 11 (0.0854 at 186 s) at 0.00652 s. Whoever needs rel-L2 < 0.03 still needs the solver; whoever needs hundreds of maps at ≈ 0.05–0.09 does not. A solver-free baseline from the window geometry (Appendix A), $\hat{C} = k\, r\, \bar{s}_r\, (g/\bar{g}_r)^{\beta}$ with disc means over the radius r and k, β fitted once at S, reaches 0.2330 at M and 0.2573 at L on one CPU core: learning cuts its error 4.6-fold and 3.0-fold.

Scale transfer. Models trained at L were evaluated at XL (1024^2) and XXL (2048^2) without fine-tuning, with the official configuration (zero-shot, three seeds) and with a variant whose training target is divided by a factor computed from the inputs and the evaluation tier only—the Omniscape radius ratio r_L/r_{tier} for T4, $\sqrt{N_{\text{valid}}/512^2}$ for T1 (scale-aware; three seeds for the U-Net on T4 and for the FNO on T4 at XL, seed 1 otherwise); the ViT's interpolated positional embedding stops at XL (Figure 7). Zero-shot transfer fails on magnitude in a regular way for T4: every model lands at rel-L2 0.498–0.538 at XL and 0.739–0.764 at XXL, which equals $1 - r_L/r_{\text{tier}}$, the share of the current that the larger Omniscape radius adds and an L-trained model cannot know about (Appendix A: 0.50 and 0.75). Removing that factor recovers T4 to 0.158–0.308 at XL and 0.197–0.438 at XXL, still above in-tier errors but no longer set by a known factor; the FNO transfers best. A U-Net T4 trained *at* XL on its 880 training landscapes reaches 0.249, between the zero-shot (0.515) and scale-aware (0.194) L-trained models. For T1 the scale-aware target helps only the FNO at XXL (1.352 $\rightarrow$ 0.762). Rankings survive what magnitudes do not: the zero-shot T4 U-Net keeps Spearman 0.992 / 0.946 / 0.904 at L / XL / XXL while its rel-L2 goes 0.081 $\rightarrow$ 0.515 $\rightarrow$ 0.749.

Out-of-distribution degradation. Figure 6 gives rel-L2 on the in-tier OOD splits relative to `test_id` (values in Appendix B). Pooled, a held-out table and contrast cost little at the training tier (`test_ood` ratio 0.920–1.191 except the ViT on T1 at L, 1.485), but the pool hides opposite effects (Table 10): its held-out-table part (real tiles, 83–84 %) is *easier* than `test_id` (0.69–0.96) and its contrast-10^6 part 1.23 to 4.17 times harder (T4: 1.82–3.29), the largest in-tier degradation we measured. Published real resistance surfaces degrade T4 by 1.300–1.530 (S only; MgNO 1.763) while T1 and T3 stay within 0.941–1.147 of `test_id`—the number to quote beside the headline. The held-out-region split is easier than `test_id` everywhere (0.652–0.982): it is real-tile only, a spatial-leakage control rather than a stress test.

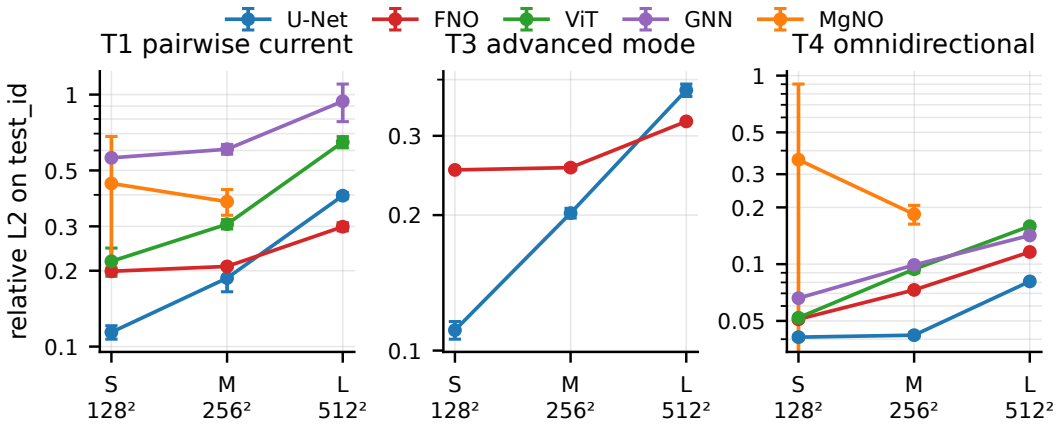


Figure 4: Relative L2 on `test_id` versus training tier per task and model (mean $\pm$ std over three seeds; MgNO's diverged seeds included). The family ordering differs between the globally coupled T1/T3 and the locally windowed T4.

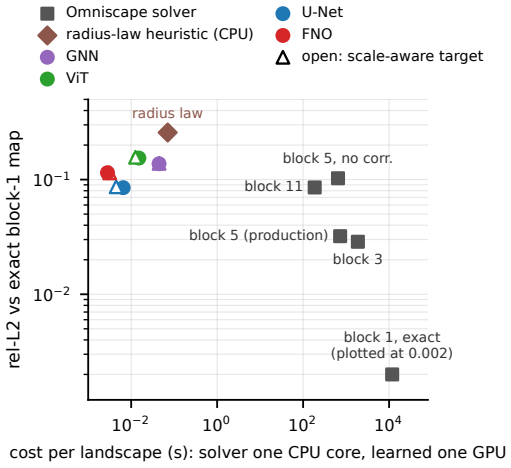


Figure 5: T4 at L: error against the exact block-1 map on the `test_id` reference subset versus cost per landscape (solver rows and the solver-free radius-law heuristic: one CPU core; learned rows: GPU inference, seeds averaged). Block 1 has error 0 and is plotted at 0.002.

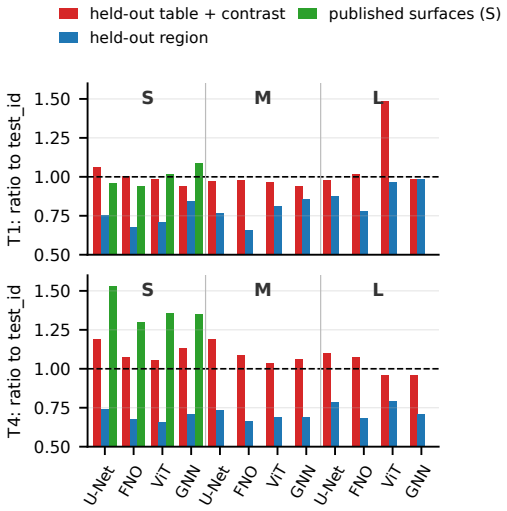


Figure 6: Per-split OOD degradation at the training tier for T1 and T4 (T3 in Appendix B): rel-L2 on each split divided by rel-L2 on `test_id` (mean over three seeds). Ratios above 1 are degradation; the held-out region split is easier by composition.

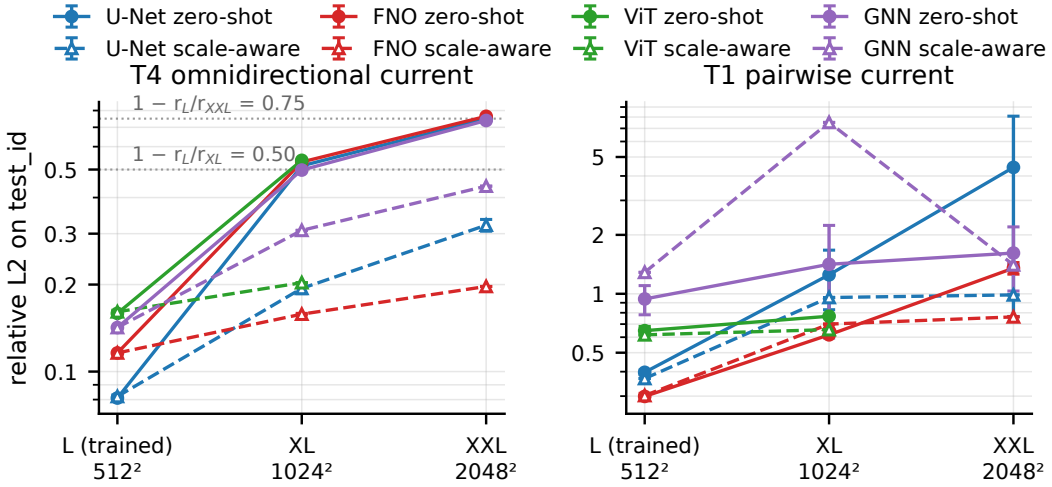


Figure 7: Scale transfer of L-trained models to XL and XXL on `test_id`: zero-shot (solid, three seeds) versus the scale-aware target (dashed; three seeds for the U-Net on T4 and the FNO on T4 at XL, seed 1 otherwise). The dotted lines in the T4 panel mark $1 - r_L/r_{\text{tier}}$, the magnitude the larger Omniscape window adds.

Sources the models never saw. T4 training sources always follow one rule (inverse resistance above its median), so a model could ignore its source channel. Re-solved under three unseen rules—uniform sources, a random field independent of resistance, a sparse $(1/R)^2$ habitat core—200 S and 100 M `test_id` landscapes give the four ranked models rel-L2 0.094–0.163 at S and 0.123–0.218 at M (1.5 to 4.6 times in-distribution; Spearman ≥ 0.77), while with the production sources swapped in every model lands within 1 % of the unchanged production-rule map (Table 11): T4 measures an operator of resistance *and* sources.

Data scaling and solver acceleration. The data-scaling ablation (Table 13) shows the 30-epoch protocol at S budget-limited for the U-Net (20,000 landscapes at the official step budget give 0.103

against 0.111 for the full run) and data-limited for the FNO (0.438 $\rightarrow$ 0.198 from 1,000 landscapes to the full set). Predicted voltages do not shorten the exact solve: warm-starting the reference AMG-PCG (relative residual 10^{-6}) from U-Net and FNO voltage heads saves no iteration on any `test_id` or `test_ood` system at S, M or L (T3 `test_id`, U-Net: 10 $\rightarrow$ 10 at S, 17 $\rightarrow$ 19 at L), because their error is high-frequency (relative Kirchhoff residual 82.5 to 3.87×10^3 times that of the zero vector). Nor is the map competitive with the solver itself on the same landscapes and targets (Appendix C; T1 over every pair): a converged AMG-PCG solve costs 0.622 / 2.961 / 16.561 s per T1 landscape and 0.045 / 0.280 / 1.599 s per T3 landscape on one core, and two iterations already beat every learned model on the median landscape of both tasks at every tier—though not on the mean, because truncated iterates are erratic on high-contrast landscapes. T1 and T3 are therefore stress tests of operator learning—global coupling through one sparse solve, singular point sources, contrast to 10^6—not compute problems; the compute problem is T4.

8 Many-query demonstration

The case for a T4 surrogate is the study that needs many maps per landscape. We ran the best T4 model at L (U-Net, seed 1; rel-L2 0.080 on `test_id`) on 20 held-out real L tiles $\times$ 8 resistance tables and compared the study-level conclusions with the solver's (Appendix H). Every conclusion is reproduced: the stability of the top-5 % regions across tables is 0.458 against the solver's 0.463 (mean pairwise IoU), the consensus core (top 5 % in $\geq$ 75 % of tables) overlaps at IoU 0.825, the ranking of tables by their effect has Spearman 0.991 with the same most-influential table on every tile, and persistent pinch points ($\geq$ 50 % of tables) are recovered with recall 0.941 but *precision* 0.657 at 3 px—at 11.8 s on one GPU against 36.3 CPU-h on one core, i.e. 11077$\times$ after training once (5,955$\times$ against the 24-thread solver; the 1.55 GPU-h of training are paid back after 13 maps). Most of that gap is numerical: bf16 rounding adds spurious local maxima (12,840 persistent candidates per tile against the solver's 5,587), and the same model in fp32 (3.71 s for the 160 maps) reaches precision 0.791 and recall 0.858, and 0.652–0.827 precision over the decision thresholds (Appendix H). Solving 6 of the 8 tables per tile lifts precision to 0.962 but costs 1.37 of the 1.82 solver CPU-h: the surrogate screens, the solver decides.

9 Limitations, ethics and maintenance

Limitations. (1) Resistance tables are structural proxies, not ecological truth; a faster approximation does not reduce resistance-surface uncertainty. (2) Learned models degrade on published surfaces, at contrast beyond the training range and across scale; final decision maps should be solver-verified, and decision metrics built on local maxima need fp32 inference. (3) T4 targets carry a quantified blocking approximation (mean rel-L2 0.029 at M, 0.030 at L). (4) Pixel size, raster size and Omniscape radius are confounded in the scale split, T4 scale transfer crosses an operator change (block 5 $\rightarrow$ 11 $\rightarrow$ 25), and only the strict scale subset isolates spatial novelty. (5) The data-scaling ablation and the demonstration are single-seed; the scale-aware variants have three seeds only for the U-Net and FNO on T4. (6) Domain-metric thresholds are not validated against ecological outcomes. (7) Not yet tested: training at XL beyond one U-Net T4 run, other T4 source rules at L and beyond, fixed-step training at M/L, MgNO at L or with fp32 / a lower learning rate (its divergences); no exact T4 reference exists at XL/XXL, so errors there are against the production target.

We do not claim that Circuitscape is slow, that exact solvers are not differentiable [49, 5], or that where to intervene is unstudied [34, 37, 8, 16].

Ethics, intended use, maintenance. No personal data (public land-cover and terrain products at $\geq$ 100 m; no protected-area polygons); every stored covariate's licence permits redistribution of derived rasters (Appendix K). Intended use is screening, not final decisions. Maintenance: versioned Hub revisions, changelog and decision log, scripted regeneration from the frozen pipeline tag, CI, GitHub issues, a DOI at acceptance.

Reproducibility. Data on the Hugging Face Hub (`Xirro/AmpScape`, CC BY 4.0) with per-tier indexes, split lists, Croissant metadata and per-sample solver provenance; code (MIT) with a locked environment and every generation, training, evaluation and leaderboard script; every figure, table and number is regenerated or checked by script from the published result files. Compute: 16 552 CPU core-hours (generation, precision pass) and 717.4 GPU-hours (800 jobs, all learning).

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A Why zero-shot T4 scale transfer fails by $1 - r_{\text{train}}/r_{\text{eval}}$

Facts used: Omniscape window radius per tier r = 16, 32, 64, 128 and 256 px for S, M, L, XL and XXL with block size b = largest odd integer $\leq r/10$; source strength $s(x) \in [0, 1]$ = inverse-resistance suitability above its 0.5 quantile rescaled to a maximum of 1 per landscape; targets are cumulative current summed over windows; measured zero-shot errors of the L-trained models on `test_id`: rel-L2 0.515 / 0.531 / 0.538 / 0.498 (U-Net / FNO / ViT / GNN) at XL and 0.749 / 0.764 / 0.739 (U-Net / FNO / GNN) at XXL, three seeds each with std ≤ 0.003.

Magnitude scales with the window radius. Omniscape visits every block centre t and solves one circuit per visit: every pixel x within distance r of t injects a current proportional to $s(x)$ and t is grounded; the per-visit maps are summed into the cumulative current C. (1) The injected current per visit is $\sum_{|x-t|\leq r} s(x) \approx \bar{s}\,\pi r^2$, with $\bar{s}$ the mean source strength of the window (an input the model sees, independent of the tier). (2) Within one visit the injected current flows towards the single grounded target; for a pixel at distance ρ from t the current crossing a front of length $\propto \rho$ is $\propto \bar{s}\rho^2/\rho = \bar{s}\rho$, of order $\bar{s}r$ over most of the window; the resistance-dependent proportionality constant is what the surrogate learns and its dependence on r is the same for every pixel. (3) The number of visits covering a pixel is the number of block centres within distance r, $\propto (r/b)^2$, a constant (≈ 100) under the block rule. Hence $C(y) \propto \bar{s} \cdot r \cdot g(y)$, where g collects the resistance-dependent, tier-independent factors: doubling the radius doubles the cumulative current to first order.

The zero-shot error. A model trained at L learns $C_L \propto r_L\,\bar{s}\,g$. Evaluated at radius r_{eval} on a landscape with the same resistance statistics it predicts $P \approx C_{\text{eval}}\,(r_L/r_{\text{eval}})$: the same g, the wrong magnitude. With $\kappa = r_L/r_{\text{eval}}$,

$$\text{rel-L2} = \frac{\|P - C_{\text{eval}}\|_2}{\|C_{\text{eval}}\|_2} = \frac{\|(\kappa - 1)C_{\text{eval}} + \varepsilon\|_2}{\|C_{\text{eval}}\|_2} \approx 1 - \kappa,$$

where ε is the model's intrinsic error (its in-tier rel-L2 is 0.081–0.159) and the approximation holds when $(1 - \kappa)\|C\|$ dominates $\|\varepsilon\|$. Predicted: XL $1 - 64/128 = 0.50$, XXL $1 - 64/256 = 0.75$; measured 0.498–0.538 and 0.739–0.764, the small excess accounted for by ε in quadrature. The near-identical values across four architectures with std ≤ 0.003 across seeds are consistent with an error set by the operator, not the model. Rank metrics ignore a common factor, which is why Spearman stays at 0.894–0.958 at XL while rel-L2 collapses.

The scale-aware target. The variant trains on $k{\cdot}C$ with $k = r_{\text{ref}}/r_{\text{tier}}$, $r_{\text{ref}} = 64$, so the regression target $k \cdot C \propto r_{\text{ref}}\,\bar{s}\,g$ is tier-independent; at evaluation the prediction is divided by k_{eval}, restoring $C_{\text{eval}} \propto r_{\text{eval}}\,\bar{s}\,g$. No quantity of the evaluation tier other than its radius (a protocol constant) enters, and no XL/XXL data is used in training or in the inverse. What remains is the error that is not a common factor: 0.158–0.308 at XL and 0.197–0.438 at XXL against in-tier 0.081–0.159—the operator-learning transfer error proper, since b and r both scale and the resistance field has a different pixel size and extent. For a pairwise solve a unit current is injected per source regardless of landscape size; under a uniform rescaling by s per axis the current crossing any front spreads over $\propto s$ pixels, so the per-pixel current $\propto 1/s$ while the valid pixel count $N \propto s^2$; $k = \sqrt{N_{\text{valid}}/N_{\text{ref}}}$ with $N_{\text{ref}} = 512^2$ makes $k \cdot C$ invariant and is computed from the input mask alone. The measured throughput error of the zero-shot T1 transfers (0.434–0.985 at XL) is the signature of the same mismatch; the correction helps the FNO at XXL (1.352 → 0.762) but not the U-Net (0.957 / 0.987 at XL / XXL), whose receptive field does not cover the longer flow paths of larger landscapes. The argument assumes b/r fixed by the block rule, sources with tier-independent mean strength, the same resistance statistics across tiers (real XL/XXL tiles have coarser pixels than L, part of the residual) and windows that fit inside the landscape for most pixels; it explains the magnitude of the zero-shot error and does not predict which model transfers best once the magnitude is corrected.

B Full result tables

Every table is generated by `scripts/paper_tables_tex.py` from the regenerated tables under `paper/tables/` and `docs/tables/`; the source file is a comment in each table's `.tex`.

Table 3: Learned baselines on `test_id`, every metric (mean ± std over three seeds; median speed-up = solver wall time / inference time per landscape; GPU-h = training cost per run).

task	tier	model	rel-L2 [median]	MAE $\log_{10}$	top-5 % IoU	pinch recall	Spearman	speed-up	GPU-h / run
T1	S	U-Net	0.114 ± 0.007 [0.111]	0.042 ± 0.002	0.858 ± 0.005	0.915 ± 0.001	0.993 ± 0.000	625	0.84
T1	S	FNO	0.199 ± 0.002 [0.199]	0.111 ± 0.001	0.723 ± 0.004	0.814 ± 0.001	0.967 ± 0.000	503	0.96
T1	S	ViT	0.218 ± 0.028 [0.229]	0.120 ± 0.027	0.576 ± 0.066	0.793 ± 0.027	0.950 ± 0.018	550	0.80
T1	S	GNN	0.561 ± 0.005 [0.563]	0.256 ± 0.002	0.547 ± 0.001	0.711 ± 0.004	0.891 ± 0.001	48	5.06
T1	S	MgNO	0.444 ± 0.238 [0.419]	0.152 ± 0.133	0.645 ± 0.220	0.771 ± 0.132	0.939 ± 0.074	488	0.53
T1	M	U-Net	0.187 ± 0.022 [0.187]	0.091 ± 0.008	0.751 ± 0.029	0.790 ± 0.042	0.972 ± 0.005	537	0.91
T1	M	FNO	0.208 ± 0.003 [0.209]	0.115 ± 0.001	0.723 ± 0.002	0.728 ± 0.001	0.964 ± 0.000	979	1.28
T1	M	ViT	0.306 ± 0.013 [0.299]	0.127 ± 0.003	0.546 ± 0.004	0.679 ± 0.004	0.945 ± 0.001	1115	0.50
T1	M	GNN	0.607 ± 0.027 [0.600]	0.282 ± 0.002	0.514 ± 0.003	0.527 ± 0.002	0.877 ± 0.002	75	6.10
T1	M	MgNO	0.376 ± 0.044 [0.389]	0.115 ± 0.028	0.679 ± 0.050	0.737 ± 0.039	0.964 ± 0.011	792	0.85
T1	L	U-Net	0.397 ± 0.011 [0.392]	0.200 ± 0.008	0.549 ± 0.023	0.500 ± 0.009	0.876 ± 0.008	691	1.49
T1	L	FNO	0.299 ± 0.011 [0.297]	0.141 ± 0.001	0.665 ± 0.003	0.531 ± 0.003	0.949 ± 0.001	1333	0.89
T1	L	ViT	0.648 ± 0.033 [0.644]	0.375 ± 0.002	0.146 ± 0.001	0.472 ± 0.009	0.665 ± 0.000	277	4.87
T1	L	GNN	0.941 ± 0.160 [0.925]	0.382 ± 0.013	0.501 ± 0.008	0.387 ± 0.018	0.861 ± 0.003	87	8.34
T3	S	U-Net	0.111 ± 0.005 [0.110]	0.051 ± 0.001	0.827 ± 0.001	0.890 ± 0.005	0.989 ± 0.000	319	0.72
T3	S	FNO	0.252 ± 0.001 [0.252]	0.147 ± 0.002	0.639 ± 0.001	0.695 ± 0.008	0.929 ± 0.002	219	0.92
T3	M	U-Net	0.202 ± 0.005 [0.202]	0.106 ± 0.002	0.715 ± 0.006	0.766 ± 0.004	0.959 ± 0.002	223	1.18
T3	M	FNO	0.255 ± 0.003 [0.254]	0.149 ± 0.001	0.647 ± 0.004	0.656 ± 0.002	0.931 ± 0.001	125	2.99
T3	L	U-Net	0.379 ± 0.012 [0.374]	0.202 ± 0.003	0.522 ± 0.013	0.557 ± 0.031	0.859 ± 0.004	303	1.54
T3	L	FNO	0.323 ± 0.002 [0.323]	0.182 ± 0.001	0.569 ± 0.003	0.497 ± 0.001	0.902 ± 0.001	654	0.91
T4	S	U-Net	0.041 ± 0.001 [0.041]	0.022 ± 0.000	0.861 ± 0.002	0.918 ± 0.003	0.996 ± 0.000	167020	0.79
T4	S	FNO	0.051 ± 0.001 [0.051]	0.038 ± 0.001	0.820 ± 0.000	0.862 ± 0.002	0.991 ± 0.000	124105	1.01
T4	S	ViT	0.052 ± 0.001 [0.052]	0.032 ± 0.001	0.811 ± 0.006	0.893 ± 0.005	0.992 ± 0.000	106498	0.85
T4	S	GNN	0.066 ± 0.000 [0.066]	0.038 ± 0.000	0.742 ± 0.000	0.840 ± 0.004	0.982 ± 0.000	12545	5.18
T4	S	MgNO	0.358 ± 0.544 [0.045]	0.385 ± 0.617	0.563 ± 0.471	0.627 ± 0.479	0.993 ± 0.000	141008	0.52
T4	M	U-Net	0.042 ± 0.001 [0.041]	0.022 ± 0.001	0.867 ± 0.016	0.914 ± 0.003	0.996 ± 0.000	32789	1.97
T4	M	FNO	0.073 ± 0.000 [0.073]	0.049 ± 0.000	0.794 ± 0.001	0.768 ± 0.007	0.986 ± 0.000	34757	3.13
T4	M	ViT	0.094 ± 0.004 [0.093]	0.056 ± 0.005	0.720 ± 0.014	0.819 ± 0.008	0.979 ± 0.002	51358	1.82
T4	M	GNN	0.099 ± 0.001 [0.099]	0.054 ± 0.000	0.698 ± 0.001	0.775 ± 0.005	0.972 ± 0.000	6604	9.44
T4	M	MgNO	0.184 ± 0.021 [0.189]	0.149 ± 0.014	0.592 ± 0.035	0.725 ± 0.042	0.949 ± 0.012	123357	0.25
T4	L	U-Net	0.081 ± 0.001 [0.081]	0.038 ± 0.001	0.812 ± 0.005	0.825 ± 0.009	0.992 ± 0.000	111998	1.55
T4	L	FNO	0.116 ± 0.000 [0.116]	0.070 ± 0.000	0.723 ± 0.002	0.559 ± 0.011	0.975 ± 0.000	245969	0.91
T4	L	ViT	0.159 ± 0.004 [0.161]	0.097 ± 0.003	0.605 ± 0.009	0.735 ± 0.010	0.953 ± 0.002	49239	5.12
T4	L	GNN	0.142 ± 0.001 [0.142]	0.071 ± 0.000	0.641 ± 0.003	0.694 ± 0.007	0.962 ± 0.000	15910	9.29

Table 4: Configuration selection per model × task × tier on validation loss alone (masked MSE in the log-target space; mean over three pre-tuning seeds ± std against the mean of three tuned seeds; the tuned configuration is official when it beats the pre-tuning mean by more than std/$\sqrt{3}$).

tier	task	model	variant	val loss pre-tuning	tuned	Δ	official
M	T1	FNO	w48	0.0495 ± 0.0003	0.0461	-6.9 %	tuned (w48)
M	T1	GNN	ms8	0.2156 ± 0.0008	0.1603	-25.6 %	tuned (ms8)
M	T1	U-Net	w64	0.0295 ± 0.0060	0.0239	-19.0 %	tuned (w64)
M	T1	ViT	p16g16	0.1336 ± 0.0820	0.0504	-62.3 %	tuned (p16g16)
M	T3	FNO	w48	0.0630 ± 0.0020	0.0555	-11.9 %	tuned (w48)
M	T3	U-Net	w64	0.0276 ± 0.0010	0.0272	-1.4 %	pre-tuning
M	T4	FNO	w48	0.0118 ± 0.0001	0.0110	-6.5 %	tuned (w48)
M	T4	GNN	ms8	0.0120 ± 0.0002	0.0129	+8.0 %	pre-tuning
M	T4	U-Net	w64	0.0035 ± 0.0001	0.0031	-12.5 %	tuned (w64)
M	T4	ViT	p16g16	0.0134 ± 0.0024	0.0157	+16.8 %	pre-tuning
L	T1	FNO	w48	0.0626 ± 0.0003	0.0600	-4.1 %	tuned (w48)
L	T1	GNN	ms8	0.3204 ± 0.0068	0.2693	-15.9 %	tuned (ms8)
L	T3	FNO	w48	0.0863 ± 0.0028	0.0789	-8.6 %	tuned (w48)
L	T4	FNO	w48	0.0212 ± 0.0004	0.0206	-2.7 %	tuned (w48)
L	T4	GNN	ms8	0.0220 ± 0.0001	0.0202	-8.3 %	tuned (ms8)

Table 5: Effect of the per-tier tuning on `test_id` (mean $\pm$ std over three seeds), with the official choice of Table 4. The validation rule keeps the pre-tuning configuration for U-Net T3 M despite a better test rel-L2 and for ViT and GNN T4 M where the tuned one is worse on test: test metrics are never read.

task	tier	model	official	rel-L2 pre-tuning $\rightarrow$ tuned	Δ	MAE $\log_{10}$ pre-tuning $\rightarrow$ tuned	Δ
T1	L	FNO	tuned	$0.361 \pm 0.024 \rightarrow 0.299 \pm 0.012$	-17.3 %	$0.146 \pm 0.001 \rightarrow 0.141 \pm 0.001$	-3.7 %
T1	L	GNN	tuned	$1.067 \pm 0.026 \rightarrow 0.941 \pm 0.160$	-11.8 %	$0.416 \pm 0.001 \rightarrow 0.382 \pm 0.012$	-8.2 %
T1	M	FNO	tuned	$0.248 \pm 0.016 \rightarrow 0.208 \pm 0.003$	-16.1 %	$0.122 \pm 0.001 \rightarrow 0.115 \pm 0.001$	-6.0 %
T1	M	GNN	tuned	$0.849 \pm 0.005 \rightarrow 0.607 \pm 0.027$	-28.5 %	$0.337 \pm 0.002 \rightarrow 0.282 \pm 0.003$	-16.5 %
T1	M	U-Net	tuned	$0.212 \pm 0.023 \rightarrow 0.187 \pm 0.022$	-11.7 %	$0.103 \pm 0.014 \rightarrow 0.091 \pm 0.009$	-12.0 %
T1	M	ViT	tuned	$0.419 \pm 0.123 \rightarrow 0.306 \pm 0.012$	-27.0 %	$0.240 \pm 0.099 \rightarrow 0.127 \pm 0.003$	-46.9 %
T3	L	FNO	tuned	$0.347 \pm 0.004 \rightarrow 0.323 \pm 0.002$	-6.8 %	$0.195 \pm 0.004 \rightarrow 0.182 \pm 0.001$	-6.5 %
T3	M	FNO	tuned	$0.284 \pm 0.006 \rightarrow 0.255 \pm 0.003$	-10.2 %	$0.163 \pm 0.004 \rightarrow 0.149 \pm 0.001$	-9.0 %
T3	M	U-Net	pre-tuning	$0.202 \pm 0.006 \rightarrow 0.192 \pm 0.009$	-5.0 %	$0.106 \pm 0.002 \rightarrow 0.105 \pm 0.006$	-1.6 %
T4	L	FNO	tuned	$0.120 \pm 0.001 \rightarrow 0.116 \pm 0.000$	-3.3 %	$0.073 \pm 0.001 \rightarrow 0.070 \pm 0.000$	-3.7 %
T4	L	GNN	tuned	$0.142 \pm 0.001 \rightarrow 0.142 \pm 0.001$	+0.0 %	$0.074 \pm 0.000 \rightarrow 0.071 \pm 0.001$	-4.5 %
T4	M	FNO	tuned	$0.079 \pm 0.001 \rightarrow 0.073 \pm 0.001$	-7.6 %	$0.052 \pm 0.001 \rightarrow 0.049 \pm 0.000$	-6.4 %
T4	M	GNN	pre-tuning	$0.099 \pm 0.001 \rightarrow 0.103 \pm 0.001$	+4.7 %	$0.054 \pm 0.001 \rightarrow 0.055 \pm 0.000$	+2.5 %
T4	M	U-Net	tuned	$0.051 \pm 0.000 \rightarrow 0.042 \pm 0.001$	-18.3 %	$0.025 \pm 0.000 \rightarrow 0.022 \pm 0.001$	-13.3 %
T4	M	ViT	pre-tuning	$0.094 \pm 0.004 \rightarrow 0.103 \pm 0.001$	+8.8 %	$0.056 \pm 0.005 \rightarrow 0.061 \pm 0.001$	+9.5 %

Table 6: T4 at tier M against the exact block-1 Omniscape map on the `test_id` reference subset ($n = 400$ landscapes): Omniscape solver rows (corr. = artefact correction; cost = single-core wall time per landscape) and learned models (cost = median batch-amortised GPU inference per landscape; median = median over seeds; seeds averaged otherwise).

method	cost (s)	rel-L2	median	non-source rel-L2	MAE $\log_{10}$	top-5 % IoU	pinch recall	Spearman
block 1 (exact)	776	0.0000	–	0.0000	0.0000	1.0000	1.0000	1.0000
block 3, corr. off	112	0.1105	–	0.1069	0.0260	0.6105	0.7658	0.9856
prod. block 3, corr. on	111	0.0291	–	0.0371	0.0119	0.9357	0.9312	0.9983
block 7, corr. on	25.5	0.0983	–	0.1064	0.0526	0.8081	0.8081	0.9845
block 7, corr. off	25.4	0.2918	–	0.3771	0.0679	0.5468	0.9662	0.9664
GNN (3 seeds)	0.0164	0.0928	0.0929	–	0.0511	0.7034	0.7770	0.9783
radius law, $k = 2/3$ (CPU)	0.012	0.4098	–	–	0.3152	0.3107	–	0.7519
radius law, fitted (CPU)	0.012	0.2330	–	–	0.2281	0.5022	–	0.9049
U-Net tuned (3 seeds)	0.00328	0.0510	0.0515	–	0.0235	0.8532	0.9035	0.9957
FNO tuned (3 seeds)	0.003	0.0761	0.0762	–	0.0479	0.7872	0.7530	0.9866
ViT (3 seeds)	0.00205	0.0931	0.0919	–	0.0536	0.7202	0.8145	0.9822
FNO, scale-aware target (1 seed)	0.00144	0.0813	0.0813	–	0.0511	0.7747	0.7399	0.9854
U-Net, scale-aware target (1 seed)	0.000987	0.0575	0.0575	–	0.0266	0.8413	0.8908	0.9948
MgNO (3 seeds)	0.000845	0.1816	0.1864	–	0.1422	0.5859	0.7084	0.9521

Table 7: T4 at tier L against the exact block-1 Omniscape map on the `test_id` reference subset ($n = 48$ landscapes): Omniscape solver rows (corr. = artefact correction; cost = single-core wall time per landscape) and learned models (cost = median batch-amortised GPU inference per landscape; median = median over seeds; seeds averaged otherwise).

method	cost (s)	rel-L2	median	non-source rel-L2	MAE $\log_{10}$	top-5 % IoU	pinch recall	Spearman
block 1 (exact)	1.18e+04	0.0000	–	0.0000	0.0000	1.0000	1.0000	1.0000
block 3, corr. on	1.88e+03	0.0287	–	0.0268	0.0078	0.9327	0.9331	0.9981
prod. block 5, corr. on	727	0.0321	–	0.0341	0.0125	0.9257	0.9286	0.9978
block 5, corr. off	648	0.1027	–	0.1154	0.0206	0.7382	0.9763	0.9856
block 11, corr. on	186	0.0854	–	0.0928	0.0380	0.8206	0.8143	0.9880
radius law, fitted (CPU)	0.0707	0.2573	–	–	0.2376	0.4702	–	0.8858
radius law, $k = 2/3$ (CPU)	0.0707	0.4418	–	–	0.3195	0.2536	–	0.6925
GNN tuned, scale-aware target (1 seed)	0.0446	0.1385	0.1385	–	0.0751	0.6081	0.6832	0.9581
GNN tuned (3 seeds)	0.0445	0.1375	0.1376	–	0.0755	0.6053	0.6686	0.9574
ViT (3 seeds)	0.0151	0.1541	0.1537	–	0.0960	0.5808	0.6983	0.9577
ViT, scale-aware target (1 seed)	0.0126	0.1566	0.1566	–	0.0982	0.5851	0.7129	0.9536
U-Net (3 seeds)	0.00652	0.0850	0.0849	–	0.0412	0.7894	0.8029	0.9905
U-Net, scale-aware target (3 seeds)	0.00447	0.0865	0.0870	–	0.0421	0.7851	0.7949	0.9902
FNO tuned, scale-aware target (3 seeds)	0.00317	0.1144	0.1145	–	0.0736	0.7109	0.5583	0.9776
FNO tuned (3 seeds)	0.00281	0.1148	0.1145	–	0.0736	0.7105	0.5584	0.9776

Table 8: Scale transfer on `test_id`: models trained at L evaluated at L, XL and XXL at batch 1 with the training tier's normalisation statistics; zero-shot = official configuration (three seeds), scale-aware = `--target-norm scale` (seed 1). The ViT's interpolated positional embedding stops at XL.

task	model	variant	eval tier	seeds	rel-L2 [median]	Spearman	top-5 % IoU	throughput err.
T1	FNO	scale-aware	L	1	0.302 [0.302]	0.949	0.664	0.085
T1	FNO	scale-aware	XL	1	0.700 [0.700]	0.928	0.496	0.453
T1	FNO	scale-aware	XXL	1	0.762 [0.762]	0.918	0.449	0.490
T1	FNO	zero-shot	L	3	0.299 ± 0.011 [0.297]	0.949 ± 0.001	0.665 ± 0.003	0.083 ± 0.003
T1	FNO	zero-shot	XL	3	0.617 ± 0.010 [0.611]	0.932 ± 0.000	0.554 ± 0.001	0.434 ± 0.083
T1	FNO	zero-shot	XXL	3	1.352 ± 0.092 [1.393]	0.921 ± 0.001	0.507 ± 0.003	1.689 ± 0.182
T1	GNN	scale-aware	L	1	1.287 [1.287]	0.860	0.495	0.854
T1	GNN	scale-aware	XL	1	7.509 [7.509]	0.736	0.381	0.725
T1	GNN	scale-aware	XXL	1	1.405 [1.405]	0.723	0.355	0.735
T1	GNN	zero-shot	L	3	0.941 ± 0.160 [0.925]	0.861 ± 0.003	0.501 ± 0.008	0.873 ± 0.075
T1	GNN	zero-shot	XL	3	1.415 ± 0.821 [0.978]	0.741 ± 0.001	0.377 ± 0.001	0.985 ± 0.043
T1	GNN	zero-shot	XXL	3	1.615 ± 0.582 [1.334]	0.727 ± 0.002	0.352 ± 0.007	1.709 ± 0.125
T1	U-Net	scale-aware	L	1	0.369 [0.369]	0.873	0.542	0.116
T1	U-Net	scale-aware	XL	1	0.957 [0.957]	0.729	0.413	0.973
T1	U-Net	scale-aware	XXL	1	0.987 [0.987]	0.597	0.286	0.980
T1	U-Net	zero-shot	L	3	0.397 ± 0.011 [0.392]	0.876 ± 0.008	0.549 ± 0.023	0.107 ± 0.011
T1	U-Net	zero-shot	XL	3	1.250 ± 0.421 [1.026]	0.721 ± 0.020	0.418 ± 0.008	0.943 ± 0.018
T1	U-Net	zero-shot	XXL	3	4.425 ± 3.659 [4.015]	0.461 ± 0.034	0.301 ± 0.006	0.959 ± 0.033
T1	ViT	scale-aware	L	1	0.617 [0.617]	0.665	0.147	0.329
T1	ViT	scale-aware	XL	1	0.656 [0.656]	0.657	0.141	0.363
T1	ViT	zero-shot	L	3	0.648 ± 0.033 [0.644]	0.665 ± 0.000	0.146 ± 0.001	0.354 ± 0.007
T1	ViT	zero-shot	XL	3	0.768 ± 0.006 [0.767]	0.657 ± 0.001	0.141 ± 0.001	0.591 ± 0.069
T4	FNO	scale-aware	L	3	0.116 ± 0.001 [0.116]	0.975 ± 0.000	0.723 ± 0.002	–
T4	FNO	scale-aware	XL	3	0.158 ± 0.001 [0.158]	0.958 ± 0.000	0.677 ± 0.003	–
T4	FNO	scale-aware	XXL	1	0.197 [0.197]	0.941	0.637	–
T4	FNO	zero-shot	L	3	0.116 ± 0.000 [0.116]	0.975 ± 0.000	0.723 ± 0.002	–
T4	FNO	zero-shot	XL	3	0.531 ± 0.002 [0.532]	0.958 ± 0.000	0.676 ± 0.002	–
T4	FNO	zero-shot	XXL	3	0.764 ± 0.001 [0.764]	0.942 ± 0.000	0.635 ± 0.002	–
T4	GNN	scale-aware	L	1	0.142 [0.142]	0.962	0.643	–
T4	GNN	scale-aware	XL	1	0.308 [0.308]	0.894	0.472	–
T4	GNN	scale-aware	XXL	1	0.438 [0.438]	0.872	0.438	–
T4	GNN	zero-shot	L	3	0.142 ± 0.001 [0.142]	0.962 ± 0.000	0.641 ± 0.003	–
T4	GNN	zero-shot	XL	3	0.498 ± 0.003 [0.497]	0.894 ± 0.001	0.469 ± 0.002	–
T4	GNN	zero-shot	XXL	3	0.739 ± 0.002 [0.738]	0.872 ± 0.001	0.435 ± 0.002	–
T4	U-Net	scale-aware	L	3	0.082 ± 0.001 [0.081]	0.991 ± 0.000	0.808 ± 0.002	–
T4	U-Net	scale-aware	XL	3	0.194 ± 0.008 [0.191]	0.946 ± 0.000	0.596 ± 0.004	–
T4	U-Net	scale-aware	XXL	3	0.321 ± 0.015 [0.314]	0.905 ± 0.000	0.505 ± 0.004	–
T4	U-Net	zero-shot	L	3	0.081 ± 0.001 [0.081]	0.992 ± 0.000	0.812 ± 0.005	–
T4	U-Net	zero-shot	XL	3	0.515 ± 0.002 [0.515]	0.946 ± 0.000	0.596 ± 0.002	–
T4	U-Net	zero-shot	XXL	3	0.749 ± 0.002 [0.749]	0.904 ± 0.001	0.505 ± 0.003	–
T4	ViT	scale-aware	L	1	0.161 [0.161]	0.951	0.602	–
T4	ViT	scale-aware	XL	1	0.203 [0.203]	0.935	0.572	–
T4	ViT	zero-shot	L	3	0.159 ± 0.004 [0.161]	0.953 ± 0.002	0.605 ± 0.009	–
T4	ViT	zero-shot	XL	3	0.538 ± 0.000 [0.538]	0.937 ± 0.002	0.573 ± 0.008	–

Table 9: Rel-L2 per in-tier split (mean over three seeds, median in brackets) and ratios to `test_id`; `test_ood` = held-out resistance table and contrast 10^6, `ood_region` = held-out biomes and realm, published = real tiles with published resistance surfaces (S only).

task	tier	model	test_id	test_ood	ood_region	published	test_ood / id	ood_region / id	published / id
T1	S	FNO	0.199 [0.199]	0.200 [0.201]	0.135 [0.135]	0.188 [0.188]	1.005	0.679	0.941
T1	S	GNN	0.561 [0.563]	0.529 [0.529]	0.476 [0.479]	0.610 [0.608]	0.942	0.847	1.087
T1	S	MgNO	0.444 [0.419]	0.445 [0.426]	0.427 [0.416]	0.427 [0.389]	1.003	0.961	0.961
T1	S	U-Net	0.114 [0.111]	0.121 [0.120]	0.086 [0.085]	0.109 [0.110]	1.061	0.753	0.958
T1	S	ViT	0.218 [0.229]	0.215 [0.224]	0.155 [0.163]	0.221 [0.238]	0.988	0.711	1.017
T1	M	FNO	0.208 [0.209]	0.204 [0.206]	0.138 [0.140]	–	0.981	0.661	–
T1	M	GNN	0.607 [0.600]	0.572 [0.560]	0.518 [0.520]	–	0.942	0.854	–
T1	M	MgNO	0.376 [0.389]	0.384 [0.396]	0.337 [0.364]	–	1.021	0.898	–
T1	M	U-Net	0.187 [0.187]	0.182 [0.184]	0.144 [0.144]	–	0.973	0.769	–
T1	M	ViT	0.306 [0.299]	0.295 [0.290]	0.247 [0.240]	–	0.965	0.809	–
T1	L	FNO	0.299 [0.297]	0.303 [0.305]	0.232 [0.232]	–	1.015	0.778	–
T1	L	GNN	0.941 [0.925]	0.927 [0.924]	0.923 [0.903]	–	0.985	0.982	–
T1	L	U-Net	0.397 [0.392]	0.387 [0.387]	0.348 [0.340]	–	0.976	0.877	–
T1	L	ViT	0.648 [0.644]	0.963 [0.692]	0.628 [0.622]	–	1.485	0.968	–
T3	S	FNO	0.252 [0.252]	0.235 [0.234]	0.170 [0.171]	0.289 [0.285]	0.930	0.674	1.147
T3	S	U-Net	0.111 [0.110]	0.113 [0.114]	0.072 [0.073]	0.126 [0.128]	1.022	0.652	1.138
T3	M	FNO	0.255 [0.254]	0.242 [0.241]	0.172 [0.171]	–	0.951	0.676	–
T3	M	U-Net	0.202 [0.202]	0.198 [0.201]	0.146 [0.147]	–	0.981	0.722	–
T3	L	FNO	0.323 [0.323]	0.297 [0.297]	0.250 [0.250]	–	0.920	0.773	–
T3	L	U-Net	0.379 [0.374]	0.357 [0.352]	0.321 [0.321]	–	0.942	0.848	–
T4	S	FNO	0.051 [0.051]	0.055 [0.055]	0.034 [0.034]	0.066 [0.065]	1.075	0.679	1.300
T4	S	GNN	0.066 [0.066]	0.075 [0.075]	0.047 [0.047]	0.089 [0.090]	1.134	0.709	1.351
T4	S	MgNO	0.358 [0.045]	0.393 [0.048]	0.319 [0.032]	0.632 [0.057]	1.098	0.890	1.763
T4	S	U-Net	0.041 [0.041]	0.049 [0.049]	0.031 [0.031]	0.063 [0.063]	1.189	0.742	1.530
T4	S	ViT	0.052 [0.052]	0.055 [0.055]	0.034 [0.034]	0.071 [0.071]	1.056	0.659	1.359
T4	M	FNO	0.073 [0.073]	0.079 [0.079]	0.048 [0.048]	–	1.088	0.664	–
T4	M	GNN	0.099 [0.099]	0.105 [0.105]	0.068 [0.069]	–	1.063	0.692	–
T4	M	MgNO	0.184 [0.189]	0.207 [0.208]	0.167 [0.177]	–	1.126	0.907	–
T4	M	U-Net	0.042 [0.041]	0.050 [0.050]	0.031 [0.030]	–	1.191	0.736	–
T4	M	ViT	0.094 [0.093]	0.098 [0.097]	0.065 [0.063]	–	1.039	0.687	–
T4	L	FNO	0.116 [0.116]	0.125 [0.125]	0.079 [0.079]	–	1.075	0.684	–
T4	L	GNN	0.142 [0.142]	0.136 [0.136]	0.100 [0.100]	–	0.959	0.708	–
T4	L	U-Net	0.081 [0.081]	0.089 [0.090]	0.064 [0.063]	–	1.102	0.788	–
T4	L	ViT	0.159 [0.161]	0.152 [0.154]	0.125 [0.127]	–	0.960	0.791	–

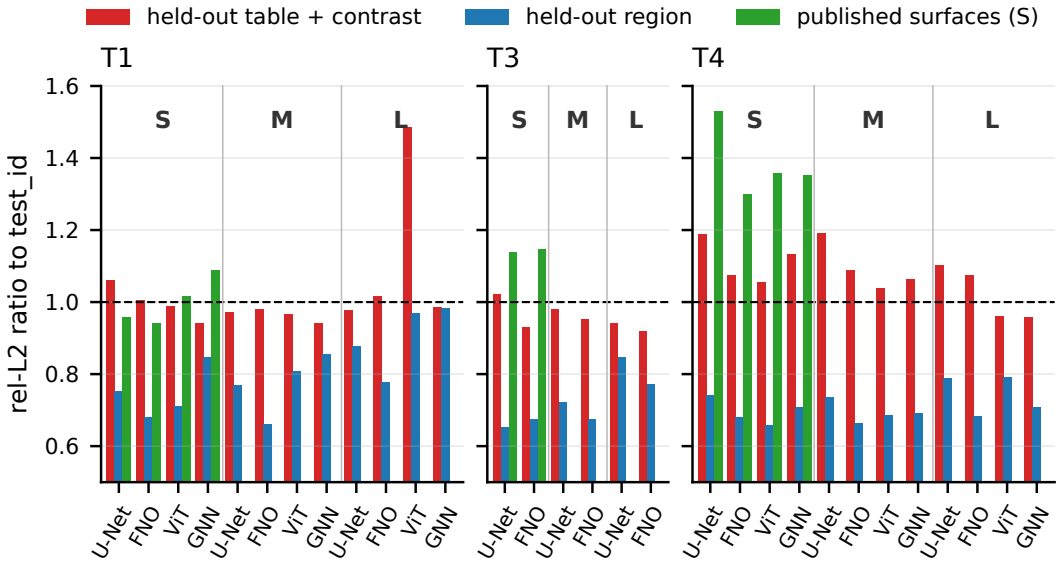


Figure 8: Per-split OOD degradation for all three tasks (the main-text Figure 6 shows T1 and T4).

Table 10: Relative L2 of the official runs by stratum (seed mean of per-seed means, as in Table 2). The interval resamples landscapes (2,000 resamples, the same for every seed; percentile interval) and so reflects landscape sampling, not seed variance (the $\pm$ of Table 2). Synthetic and real: the two families of `test_id`; contrast 10^6 and held-out table: the two parts of `test_ood`; published: `test_ood_published` (S only). Intervals for every stratum are in the source file.

task	tier	model	test_id [95 % CI]	synthetic	real	contrast 10^6	held-out table	published
T1	S	U-Net	0.114 [0.113, 0.115]	0.131	0.086	0.234	0.098	0.109
T1	S	FNO	0.199 [0.197, 0.202]	0.239	0.136	0.419	0.155	0.188
T1	S	ViT	0.218 [0.215, 0.220]	0.257	0.154	0.409	0.175	0.221
T1	S	GNN	0.561 [0.552, 0.571]	0.612	0.480	0.847	0.463	0.610
T1	S	MgNO	0.444 [0.441, 0.449]	0.453	0.429	0.547	0.424	0.427
T1	M	U-Net	0.187 [0.184, 0.190]	0.214	0.141	0.345	0.149	–
T1	M	FNO	0.208 [0.204, 0.212]	0.250	0.138	0.431	0.159	–
T1	M	ViT	0.306 [0.301, 0.310]	0.341	0.245	0.512	0.251	–
T1	M	GNN	0.607 [0.593, 0.623]	0.665	0.508	0.911	0.504	–
T1	M	MgNO	0.376 [0.373, 0.379]	0.399	0.336	0.598	0.340	–
T1	L	U-Net	0.397 [0.389, 0.405]	0.431	0.338	0.578	0.350	–
T1	L	FNO	0.299 [0.291, 0.307]	0.342	0.224	0.633	0.239	–
T1	L	ViT	0.648 [0.640, 0.657]	0.664	0.621	2.703	0.623	–
T1	L	GNN	0.941 [0.899, 0.983]	0.934	0.952	1.153	0.882	–
T3	S	U-Net	0.111 [0.107, 0.117]	0.137	0.069	0.292	0.076	0.126
T3	S	FNO	0.252 [0.249, 0.256]	0.307	0.164	0.519	0.176	0.289
T3	M	U-Net	0.202 [0.198, 0.205]	0.238	0.140	0.432	0.151	–
T3	M	FNO	0.255 [0.250, 0.260]	0.308	0.163	0.567	0.178	–
T3	L	U-Net	0.379 [0.372, 0.386]	0.416	0.315	0.562	0.317	–
T3	L	FNO	0.323 [0.315, 0.332]	0.374	0.236	0.582	0.242	–
T4	S	U-Net	0.041 [0.040, 0.042]	0.050	0.028	0.130	0.032	0.063
T4	S	FNO	0.051 [0.050, 0.052]	0.061	0.034	0.133	0.038	0.066
T4	S	ViT	0.052 [0.051, 0.053]	0.064	0.033	0.122	0.041	0.071
T4	S	GNN	0.066 [0.065, 0.067]	0.079	0.045	0.135	0.063	0.089
T4	S	MgNO	0.358 [0.352, 0.365]	0.389	0.309	0.819	0.305	0.632
T4	M	U-Net	0.042 [0.041, 0.043]	0.050	0.027	0.138	0.032	–
T4	M	FNO	0.073 [0.071, 0.075]	0.088	0.046	0.187	0.057	–
T4	M	ViT	0.094 [0.092, 0.096]	0.114	0.061	0.201	0.077	–
T4	M	GNN	0.099 [0.097, 0.101]	0.119	0.064	0.196	0.087	–
T4	M	MgNO	0.184 [0.181, 0.186]	0.196	0.163	0.309	0.186	–
T4	L	U-Net	0.081 [0.078, 0.084]	0.094	0.059	0.209	0.066	–
T4	L	FNO	0.116 [0.112, 0.121]	0.139	0.077	0.290	0.092	–
T4	L	ViT	0.159 [0.154, 0.164]	0.183	0.116	0.303	0.123	–
T4	L	GNN	0.142 [0.137, 0.147]	0.169	0.094	0.257	0.112	–

Table 11: T4 under source rules the models never saw: rel-L2 against the exact Omniscape map of each rule on the same `test_id` landscapes (200 at S, 100 at M) and Omniscape geometry. *true*: the rule's own source raster as input; *swapped*: the production source raster of the same landscape as input, so a model that ignored its source channel would score the same in both columns; production-rule map: the solver's map under the production rule used as the prediction. Rules: uniform sources on every valid pixel; a Gaussian random field independent of resistance, kept above its median; $(1/R)^2$ above its 0.8 quantile. Official runs, three seeds, fp32 inference on CPU.

tier	model	production rule	uniform: true	swapped	random field: true	swapped	habitat core: true	swapped
S	U-Net	0.037	0.113	0.778	0.159	1.240	0.126	1.460
S	FNO	0.047	0.118	0.778	0.163	1.238	0.136	1.466
S	ViT	0.047	0.106	0.779	0.112	1.237	0.108	1.460
S	GNN	0.062	0.134	0.779	0.146	1.237	0.094	1.457
S	MgNO	0.355	0.373	0.827	0.346	1.115	0.731	1.664
S	production-rule map	0	–	0.777	–	1.239	–	1.454
M	U-Net	0.038	0.123	0.813	0.176	1.115	0.161	1.946
M	FNO	0.068	0.160	0.815	0.211	1.108	0.218	1.936
M	ViT	0.088	0.159	0.816	0.148	1.107	0.166	1.936
M	GNN	0.095	0.153	0.817	0.172	1.103	0.153	1.919
M	MgNO	0.183	0.225	0.818	0.226	1.145	0.250	2.039
M	production-rule map	0	–	0.813	–	1.112	–	1.927

Table 12: Solver acceleration: AMG-preconditioned CG warm-started from the predicted voltage versus a zero start, to a relative residual of 10^{-6} (medians over systems; reduction = median of $1-$ warm/zero per system; warm-start residual = relative Kirchhoff residual of the prediction, zero start = 1). T1V = first pair of each T1 landscape.

task	tier	model	split	systems	iterations zero → warm	reduction	time zero → warm (ms)	reduction	warm-start residual
T1V	L	FNO	test_id	1928	16 → 18	-7.7 %	517.9 → 580.5	-7.5 %	34.7
T1V	L	FNO	test_ood	1278	25 → 26	-7.1 %	860.1 → 930.0	-7.3 %	91.1
T1V	M	FNO	test_id	4823	12 → 12	+0.0 %	102.5 → 105.7	-4.5 %	22.2
T1V	M	FNO	test_ood	3225	16 → 16	+0.0 %	141.3 → 143.4	-1.8 %	22.2
T1V	S	FNO	test_id	9850	9 → 10	+0.0 %	23.2 → 23.4	-0.8 %	11
T1V	S	FNO	test_ood	6460	11 → 12	+0.0 %	30.2 → 30.7	-0.2 %	11.4
T3	L	FNO	test_id	1928	17 → 18	-5.6 %	757.0 → 808.7	-6.2 %	1.62e+03
T3	L	FNO	test_ood	1275	23 → 24	-4.2 %	1159.8 → 1221.1	-4.4 %	1.89e+03
T3	M	FNO	test_id	4823	12 → 13	+0.0 %	119.9 → 120.9	-0.5 %	559
T3	M	FNO	test_ood	3222	16 → 16	+0.0 %	167.2 → 169.3	-0.5 %	653
T3	S	FNO	test_id	9850	10 → 10	+0.0 %	19.3 → 19.7	-0.1 %	181
T3	S	FNO	test_ood	6460	11 → 11	+0.0 %	25.4 → 25.6	+0.0 %	212
T1V	L	U-Net	test_id	1928	16 → 17	-6.7 %	739.5 → 821.4	-7.4 %	22.5
T1V	L	U-Net	test_ood	1278	25 → 26	-5.6 %	1248.0 → 1320.7	-5.7 %	22.5
T1V	M	U-Net	test_id	4823	12 → 12	-5.6 %	86.9 → 91.6	-6.5 %	19.4
T1V	M	U-Net	test_ood	3225	16 → 17	-4.8 %	129.4 → 137.6	-5.7 %	17.6
T1V	S	U-Net	test_id	9850	9 → 9	+0.0 %	14.4 → 14.4	+0.0 %	5.06
T1V	S	U-Net	test_ood	6460	11 → 11	+0.0 %	19.2 → 19.3	+0.1 %	4.69
T3	L	U-Net	test_id	1928	17 → 19	-9.1 %	763.2 → 858.6	-9.5 %	3.32e+03
T3	L	U-Net	test_ood	1275	23 → 25	-8.8 %	1164.0 → 1281.7	-8.8 %	3.87e+03
T3	M	U-Net	test_id	4823	12 → 13	-2.9 %	99.9 → 102.3	-1.0 %	580
T3	M	U-Net	test_ood	3222	16 → 16	-4.5 %	140.7 → 144.5	-4.5 %	783
T3	S	U-Net	test_id	9850	10 → 10	+0.0 %	19.3 → 19.1	+0.1 %	82.5
T3	S	U-Net	test_ood	6460	11 → 11	+0.0 %	25.4 → 24.5	+0.1 %	109

Table 13: Data-scaling ablation at S on T1 (seed 1): training on 1,000 / 5,000 / 20,000 landscapes for 30 epochs or for the official run's number of optimisation steps, against the official run on all training landscapes.

model	n train	regime	rel-L2	MAE $\log_{10}$	top-5 % IoU	epochs	GPU-h
FNO	1,000	fixed epochs (30)	0.575	0.255	0.416	30	0.15
FNO	1,000	fixed steps	0.438	0.243	0.462	39	0.19
FNO	5,000	fixed epochs (30)	0.357	0.180	0.577	30	0.22
FNO	5,000	fixed steps	0.298	0.173	0.597	54	0.36
FNO	20,000	fixed epochs (30)	0.243	0.137	0.665	30	0.41
FNO	20,000	fixed steps	0.246	0.138	0.664	36	0.48
FNO	61,577	official run	0.198	0.112	0.719	29	0.95
U-Net	1,000	fixed epochs (30)	0.496	0.213	0.484	30	0.17
U-Net	1,000	fixed steps	0.368	0.162	0.607	88	0.35
U-Net	5,000	fixed epochs (30)	0.316	0.155	0.604	30	0.15
U-Net	5,000	fixed steps	0.139	0.076	0.818	143	0.76
U-Net	20,000	fixed epochs (30)	0.188	0.072	0.771	30	0.31
U-Net	20,000	fixed steps	0.103	0.042	0.877	94	0.94
U-Net	61,577	official run	0.111	0.042	0.857	30	0.86

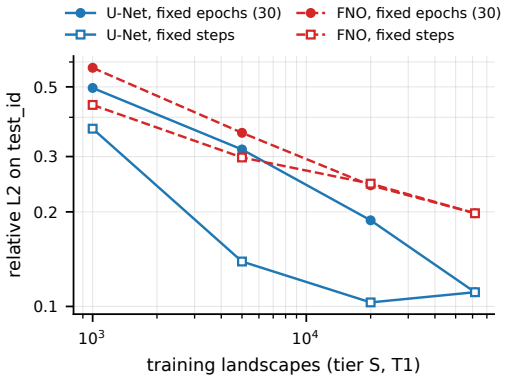


Figure 9: Data-scaling ablation at S on T1 (seed 1): relative L2 on `test_id` against the number of training landscapes in the fixed-epoch and fixed-step regimes; the rightmost point of each curve is the official run (Table 13).

C Like-for-like solver cost and cheap numerical baselines

Every solver time in the main text is the production setting under which the targets were generated: one Circuitscape or Omniscape solve per CPU core, Omniscape's own window parallelisation off. Table 14 re-times the production T4 solve with that parallelisation on; Table 15 puts the T4 surrogates on the same axis as the solver on one thread and on the node's maximum thread count, adds the training cost of one official run and gives the number of maps from which training once plus inference is cheaper in wall time than the solver. Hardware differs by construction (one GPU against one CPU node), so the table compares wall time only.

For T1 and T3 the natural competitors are not learned at all: the reference solver stopped early, and the coarsen-$\times 4$-solve-upsample baseline of Section 7. Table 16 and Figure 10 score both on a `test_id` subset per tier (the first shards in Hub order: 40 landscapes at S, 30 at M, 16 at L), with the same metrics, targets and NoData masking as the harness, and score the learned models on the same landscapes. Truncated PCG means AMG-preconditioned conjugate gradient from the zero vector on the exact graph (Circuitscape's `cg+amg` preconditioner), the current map being Circuitscape's node current of the voltage after k iterations. For T1 every pair of the landscape is solved this way (node i grounded, 1 A injected at node j; one AMG setup per grounded node) and the $K(K-1)/2$ node-current maps are summed into the cumulative map the learned models predict; run to convergence, this reproduces the stored targets to rel-L2 at most 1.3×10^{-6} on all 86 landscapes. Means and medians differ sharply: on the high-contrast landscapes that need tens of iterations per pair (up to 204), the error of a truncated iterate is not monotone in k, so a few landscapes dominate the mean of every truncated row while the median falls fast.

Table 14: Production Omniscape solve (radius / block: M 32 / 3, L 64 / 5; CHOLMOD window solver; artefact correction on) on the first `test_id` reference landscapes with Omniscape's own parallelisation (`parallelize = true`) on one CPU node; 1 thread is the production setting under which every solver time in this paper was measured.

tier	Julia threads	landscapes	wall s / landscape (median)	speed-up vs 1 thread	parallel efficiency
M	1	4	75.6	1.00×	1.00
M	8	4	46.2	1.64×	0.20
M	24	4	46.7	1.62×	0.07
L	1	2	277.0	1.00×	1.00
L	8	2	151.5	1.83×	0.23
L	24	2	149.4	1.85×	0.08

Table 15: Like-for-like cost of the T4 surrogates at M and L (rel-L2 against the exact block-1 map in Table 6 and 7): GPU inference seconds per landscape (batch-amortised, official runs, seed mean) and training GPU-hours of one run against the production solver's seconds per landscape on 1 and on 24 threads (Table 14); break-even = the number of maps from which training once plus inference costs less wall time than the solver, $N \geq T_{\text{train}}/(t_{\text{solver}} - t_{\text{inference}})$. GPU and CPU-node hours are not priced equally; only wall time is compared.

tier	model	inference s	train GPU-h	solver s (1)	speed-up (1)	solver s (24)	speed-up (24)	break-even (1)	break-even (24)
M	U-Net	0.00328	1.97	76	23,057×	46.7	14,235×	94	152
M	FNO	0.00300	3.13	76	25,208×	46.7	15,564×	149	241
L	U-Net	0.00652	1.55	277	42,487×	149.4	22,916×	20	37
L	FNO	0.00281	0.91	277	98,581×	149.4	53,172×	12	22

Table 16: Cheap numerical baselines against the learned models on the same `test_id` landscapes and the same targets (T1: the cumulative map over all pairs; T3: the advanced-mode current): rel-L2 mean / median over the landscapes. AMG-preconditioned CG from zero stopped after k iterations, with its cost in seconds per landscape on one CPU core (AMG setup included; T1: one setup per grounded node, all pairs), the cost of the full solve to relative residual 10^{-6}, coarsen-×4-solve-upsample, and the official learned model with the lowest mean (seed mean per landscape; MgNO excluded as in Table 2).

task	tier	n	PCG $k=2$	s	PCG $k=4$	PCG $k=8$	PCG to 10^{-6}: s	coarsen ×4	best learned
T1	S	40	0.344 / 0.097	0.273	0.236 / 0.005	0.069 / 0.000	0.622	0.433 / 0.420	0.137 / 0.117 (U-Net)
T1	M	30	0.297 / 0.163	0.974	0.161 / 0.017	0.069 / 0.000	2.961	0.520 / 0.570	0.211 / 0.192 (U-Net)
T1	L	16	0.361 / 0.327	3.556	0.237 / 0.136	0.128 / 0.005	16.561	0.496 / 0.526	0.424 / 0.487 (FNO)
T3	S	40	0.172 / 0.038	0.022	0.075 / 0.001	0.005 / 0.000	0.045	0.313 / 0.286	0.112 / 0.082 (U-Net)
T3	M	30	0.181 / 0.048	0.127	0.083 / 0.002	0.030 / 0.000	0.280	0.329 / 0.304	0.219 / 0.180 (U-Net)
T3	L	16	0.619 / 0.324	0.600	0.816 / 0.054	0.064 / 0.001	1.599	0.374 / 0.293	0.460 / 0.373 (FNO)

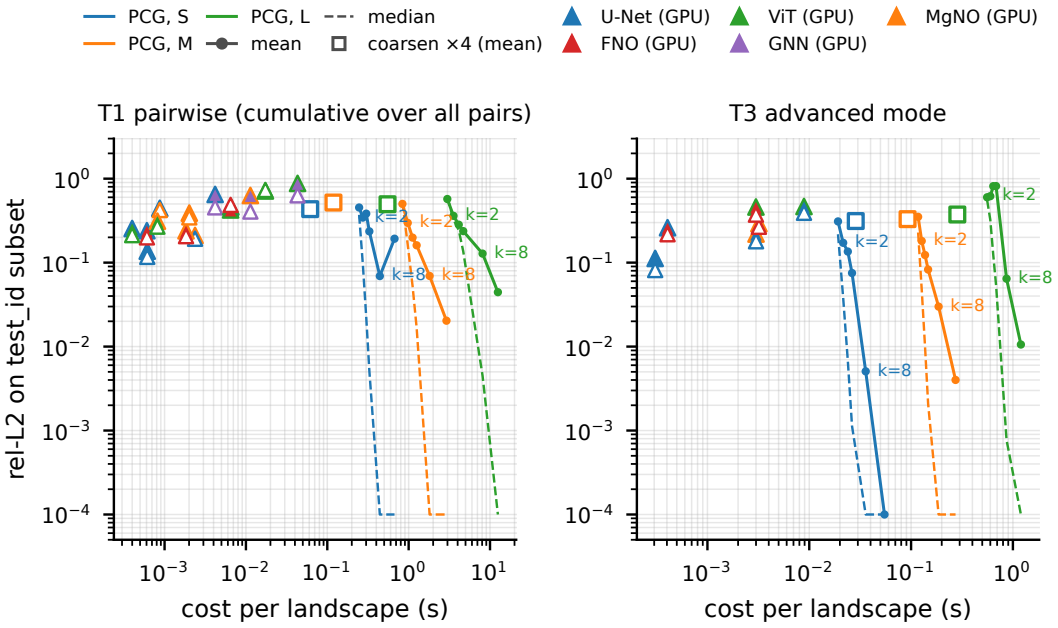
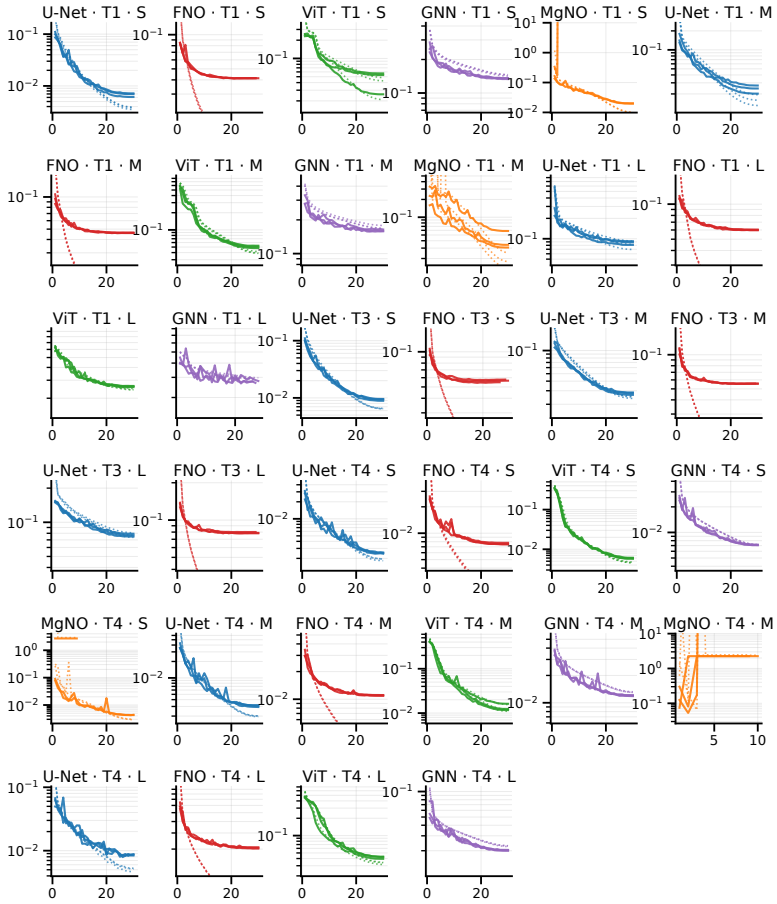


Figure 10: Error versus cost on T1 (cumulative map over all pairs) and T3, on the same `test_id` landscapes and targets: truncated AMG-PCG after k iterations (one CPU core; solid: mean, dashed: median over landscapes) and coarsen-×4 (one core; mean) against the learned models' GPU inference per landscape (triangles coloured by model; filled: mean, outlined by tier; hollow: median; official runs, seed mean per landscape). Values below 10^{-4} are plotted at 10^{-4}.

D Training and validation curves



epoch — solid: validation loss (masked MSE, log target); dotted: training loss; one line per seed (MgNO: diverged seeds included)

Figure 11: Training (dotted) and validation (solid) loss per epoch for every official run—one panel per model × task × tier, seeds overlaid (masked MSE in the log-target space; early stopping on the validation loss with patience 8; three seeds per panel; MgNO's exploding seeds are visible as flat lines).

E Dataset statistics

Contrast ($\log_{10}$, p10 / p50 / p90): synthetic 1.00 / 3.00 / 4.00 at every tier with a maximum of 10^6; real 0.76 / 1.79 / 2.79 at S rising to 2.00 / 3.00 / 3.00 at XXL with a maximum of 10^3. Real landscapes cover 14 RESOLVE biomes and 7 realms (largest: Palearctic 16,760 landscapes, Neotropic 15,405, Nearctic 12,005, Australasia 10,470). NoData fraction of real tiles: p50 0.002, p90 0.159, max 0.896.

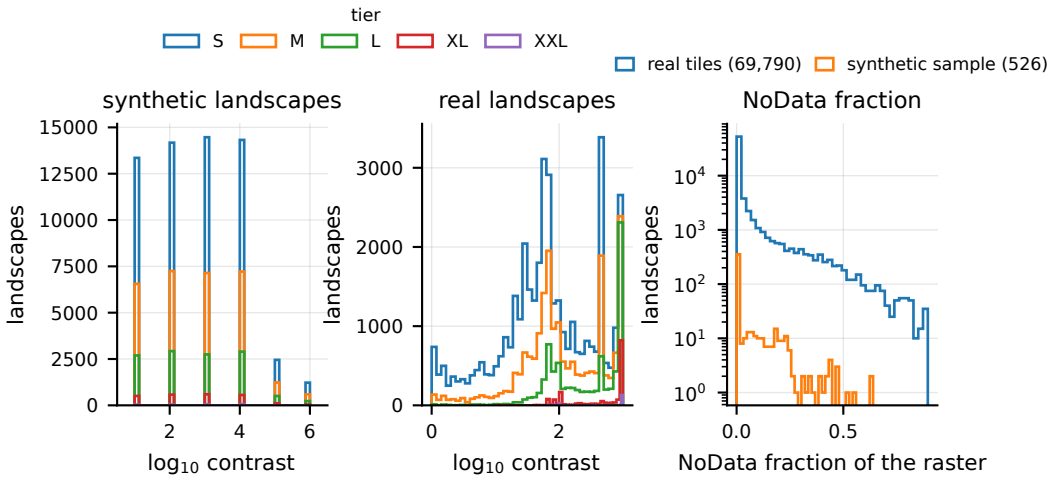


Figure 12: Dataset statistics (v1.0 index), I: contrast per tier for the synthetic and the real landscapes, and the NoData fraction of the real tiles (manifest) and of a sample of synthetic shards.

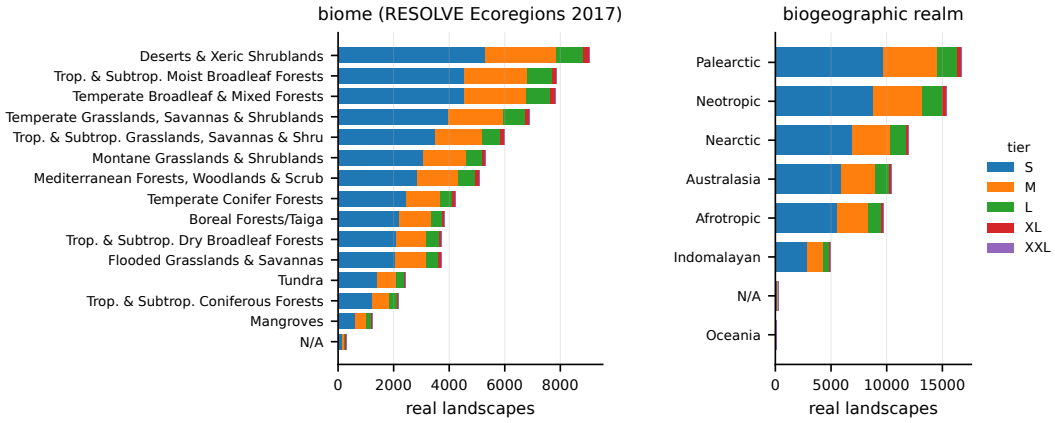


Figure 13: Dataset statistics, II: biome and biogeographic realm of the real landscapes, stacked by tier.

Table 17: Landscapes per split and tier (index counts before QC exclusion; XL after the v1.0.2 correction of amendment C3).

tier	train	val	test_id	test_ood	ood_region	QC-failing landscapes
S	63,303	10,917	9,850	6,460	9,470	1
M	31,496	5,651	4,823	3,225	4,805	4
L	12,613	2,235	1,928	1,279	1,945	10
XL	880	131	2,335	254	400	7
XXL	0	0	275	25	100	2

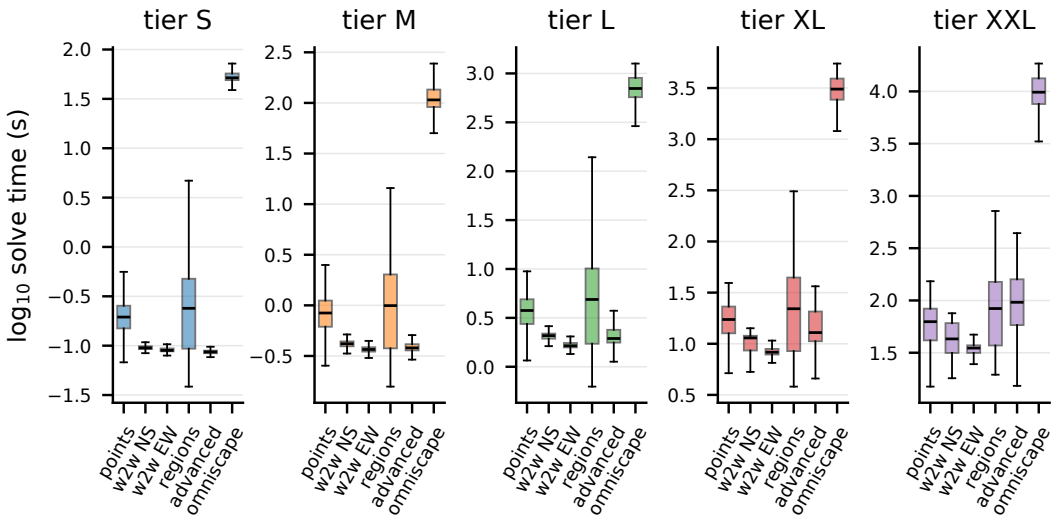


Figure 14: Dataset statistics, III: solver wall time per configuration row and tier (boxes: quartiles; whiskers: 1.5 IQR; w2w = wall-to-wall strips). Omniscape rows dominate the generation budget at every tier.

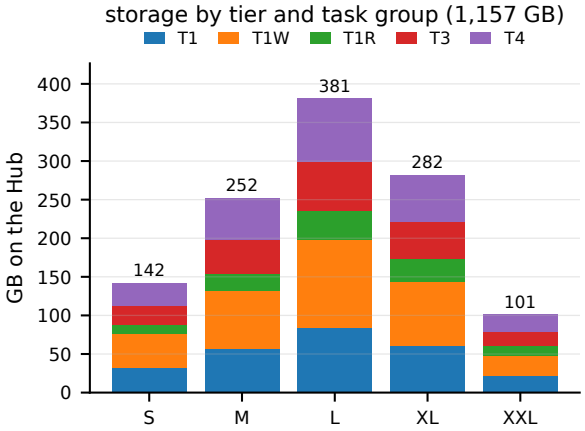


Figure 15: Dataset statistics, IV: storage on the Hub per tier and task group (GB; totals above the bars).

F Omniscape fidelity

The block rule (block = largest odd integer $\leq$ radius/10) came from a 3-sample-per-cell study against the exact block-1 map: block 3 at S deviated by 4.49 % mean / 7.60 % max relative L2 and block 5 at M by 2.00 % / 2.45 %, while coarsening XL 17→33 changed the cumulative current by 2.47 % / 2.93 % and XXL 33→65 by 9.46 % / 11.85 % (both rejected); block 1 was 6.4× (S) and 22.6× (M) more expensive at those anchors. The ≈ 1 % target was an extrapolation and proved optimistic: the production M block deviates from the exact map by 3.1 % relative L2 on average on 200 synthetic test/OOD landscapes (median 2.8 %, top-5 % IoU 0.95, pinch-point recall 0.95), and the full 1 000-sample M reference gives mean rel-L2 0.029 (median 0.027) with 4.7 % of samples above 5 % (11 % of synthetic, 1.4 % of real tiles; real tiles never above 8 %). At L (block 5) the first 60 samples gave mean 0.028 (max 0.058; 2 above 5 %); doubling the set to 120 with the same stratification gives mean 0.0303 (median 0.0273), with a heavier tail on synthetic high-contrast landscapes (max 0.2032 at contrast 10^6; 6 of 120 above 5 %, all synthetic). The Pareto rows (Table 7) and the leaderboard now use all 48 `test_id` maps; on the first 24 the U-Net had given 0.0903, on all 48 it gives 0.0850; block 1 costs a median 7.2× (M) and 19.8× (L) the production solve per landscape. T4 rows carry no single residual; each window solve is bounded by Circuitscape's enforced relative residual $\leq 10^{-4}$, with a rescue path for the contrast-10^6 solves that the stock package aborts.

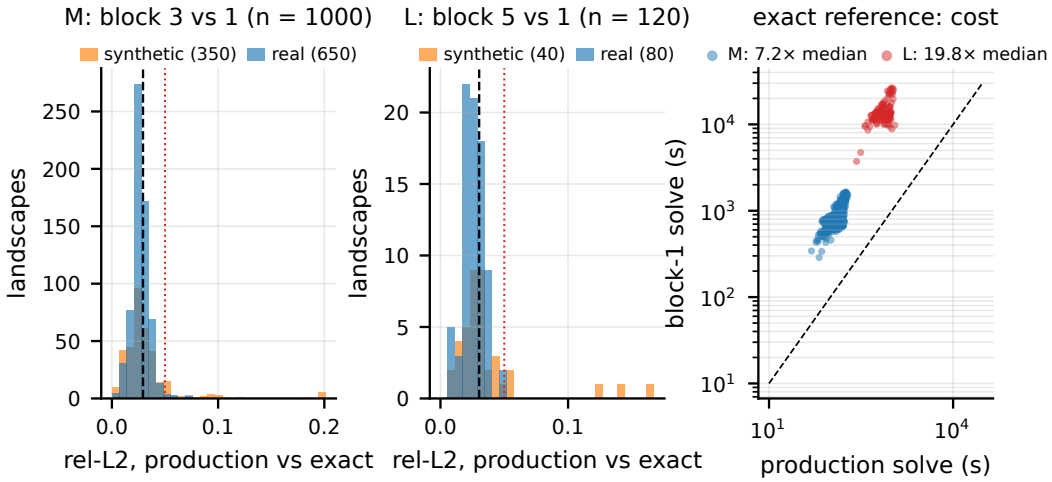


Figure 16: Fidelity of the production T4 targets on the reference subsets: per-landscape rel-L2 of the production block against the exact block-1 map at M and L (dashed: mean; dotted: the 5 % tail threshold; the few values beyond the plotted range are drawn in the last bin), and the solve-time cost of the exact reference against the production solve.

Table 18: Post-run precision pass per tier: every T1/T1W/T1R/T3 row above 10^{-9}, unmeasured, CG+AMG-solved or QC-failed was re-solved with CHOLMOD on the reduced system plus up to three refinement steps; rows that cannot reach 10^{-6} in double precision are flagged and excluded from the split lists.

tier	rows re-solved	CG+AMG → CHOLMOD	residual after: p50 / p90 / p99 / max	rows $> 10^{-9}$	rows $> 10^{-6}$ (flagged)
S	68,008	9	2.9e-13 / 3.4e-11 / 1.6e-09 / 5.0e-07	1,013	0
M	37,639	14	9.1e-13 / 1.9e-10 / 5.2e-09 / 1.4e-06	1,136	4
L	18,691	110	6.8e-12 / 4.9e-10 / 1.4e-08 / 1.5e-05	1,014	9
XL	4,774	47	6.9e-11 / 1.3e-09 / 4.3e-08 / 6.6e-06	551	5
XXL	610	9	2.7e-10 / 4.3e-09 / 2.0e-07 / 4.0e-05	147	2
total	129,722	189		3,861	20

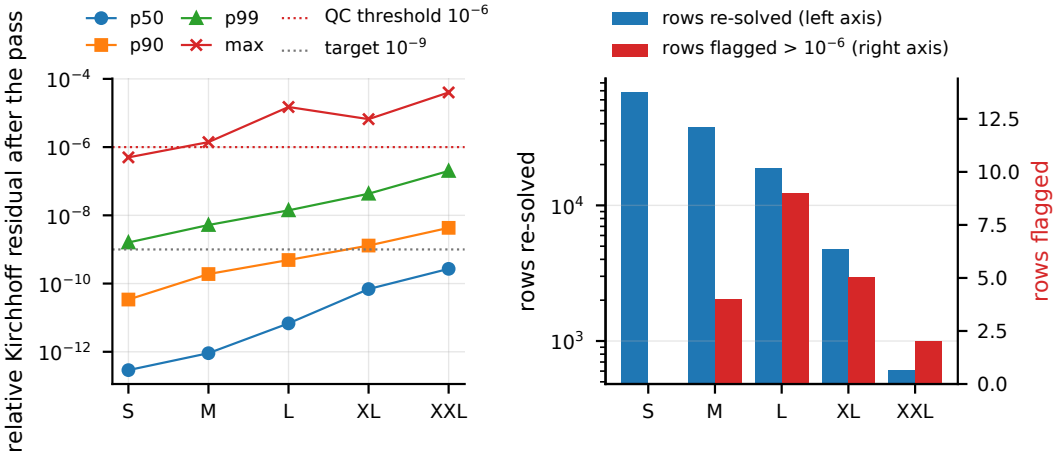


Figure 17: The post-run precision pass per tier: quantiles of the relative Kirchhoff residual after the pass against the QC threshold (10^{-6}) and the target (10^{-9}), and the number of rows re-solved and flagged.

G Precision pass

Generation ran from 2026-09-16 to 2026-09-22 followed by a 29 h precision pass, 16 552 core-hours in all (generation 14 379, precision pass 1 249, audits 30, auxiliary sets and probes 893); the eleven operational incidents and their resolutions are in the post-mortem.

H Many-query demonstration, per tile

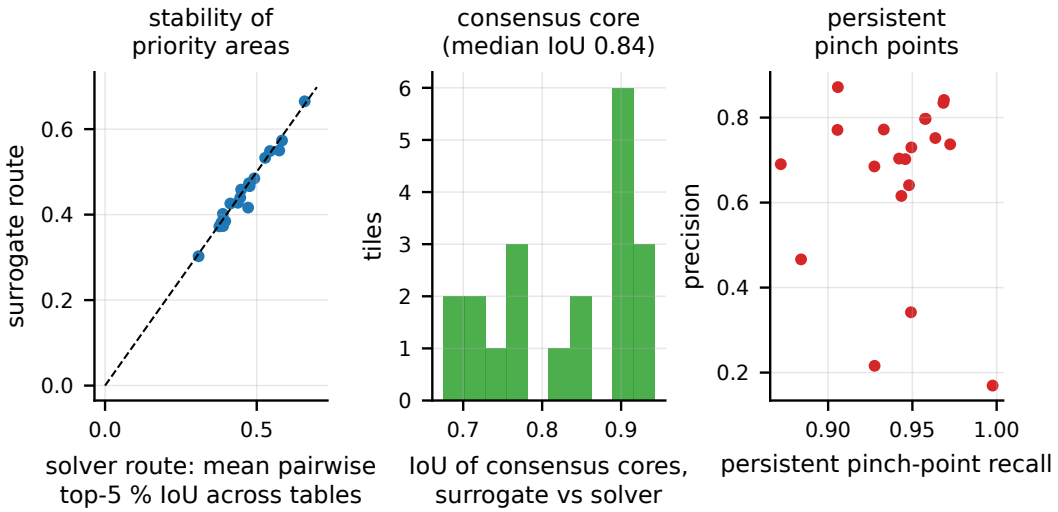


Figure 18: Per-tile agreement between the solver route and the surrogate route in the many-query demonstration (20 real L tiles × 8 resistance tables): stability of the top-5 % regions across tables, consensus-core overlap and persistent pinch-point recall/precision.

Section 8 ran the surrogate under bf16 autocast, the setting of every timing in the paper; re-run on the same GPU type (L40S) it reproduces the demonstration's precision and recall exactly. Pinch points are local maxima, and bf16 rounding of a smooth map creates extra ones: Table 19 re-runs the same model in fp32 and varies the three decision parameters, and Table 20 gives the precision a hybrid route reaches when some tables are solved exactly. Decision metrics built on local maxima should be computed from fp32 predictions.

Table 19: Persistent pinch points of the surrogate route against the solver route in the many-query demonstration (20 real L tiles $\times$ 8 tables, U-Net seed 1), over the decision parameters: a pinch point is a local maximum in the top-q % of a map; persistent = present (within 3 px) in at least the given share of the tables; matching tolerance 3 px; precision, recall and points per tile averaged over tiles. Left block: inference under bf16 autocast (the setting of Section 8); right block: fp32 inference of the same model. The demonstration's setting is $q = 5$, persistence 0.5.

top %	persistence	bf16 precision	bf16 recall	bf16 points	fp32 precision	fp32 recall	fp32 points	solver points
1	0.375	0.708	0.903	4588	0.807	0.838	2478	2440
1	0.5	0.670	0.897	3806	0.769	0.817	1982	1875
1	0.75	0.488	0.862	1743	0.652	0.738	702	640
5	0.375	0.693	0.951	14979	0.823	0.880	7502	6960
5	0.5	0.657	0.941	12840	0.791	0.858	6053	5587
5	0.75	0.520	0.911	7517	0.715	0.764	2546	2364
10	0.375	0.693	0.965	25129	0.827	0.895	12301	11212
10	0.5	0.654	0.955	21752	0.798	0.872	9956	9106
10	0.75	0.537	0.929	13542	0.727	0.803	4704	4461

Table 20: Verification budget in the many-query demonstration: m of the 8 tables per tile are solved exactly and the surrogate supplies the rest (tables in the surrogate's own order of influence, reference table first). Persistent-pinch-point precision and recall and the consensus-core IoU against the all-solver route, for bf16 (left) and fp32 (right) inference, and the solver cost per tile (one core).

tables solved	bf16 precision	bf16 recall	bf16 core IoU	fp32 precision	fp32 recall	fp32 core IoU	solver CPU-h / tile
0	0.657	0.941	0.825	0.791	0.858	0.845	0.00
2	0.669	0.941	0.883	0.796	0.863	0.897	0.44
4	0.703	0.949	0.918	0.820	0.885	0.930	0.94
5	0.833	0.964	0.929	0.909	0.914	0.941	1.16
6	0.924	0.980	0.949	0.962	0.957	0.959	1.37
7	0.973	0.991	0.965	0.988	0.980	0.971	1.59
8	1.000	1.000	1.000	1.000	1.000	1.000	1.82

I Splits per tier

The per-tier split counts are in Table 17 (Appendix E); subset sizes and the QC exclusion rule are in Section 4.

J Evaluation CLI and prediction format

`python scripts/evaluate.py --predictions <dir> --split <name>[,<name>] [--root <data root> --tier <tier> --subset <name> --out <dir> --acceleration --t4-reference <aux build> --t4-blocks <builds> --workers <n>]` writes `results.json` and `results.md`. `<dir>/predictions.h5` holds one group per sample and one subgroup per configuration with `cum_current` (float32 $H \times W$; T1, T1W, T1R, T4), optional `voltage` (float32 $H \times W$ or $P \times H \times W$), optional `pairwise_current` ($P \times H \times W$), `reff` (float64 $K \times K$; T2), `current` (float32 $H \times W$; T3) and optional `flow_potential` / `normalized` (T4), plus a group attribute `inference_time_s` per configuration (wall clock including preprocessing); `<dir>/meta.json` records model, task, tier, split, seed and notes. Only `qc_pass` rows of the requested splits are evaluated, samples absent from the file are skipped with the row count reported, processing is deterministic in sorted order, an oracle scores 0 error / 1.0 on every similarity metric and an all-zero predictor scores rel-L2 = 1 and top-q IoU = 0. The leaderboard entry schema, the validator and the submission procedure are in `docs/submission.md`.

K Licences and attributions

Every file in the source manifest (21 entries, 5.45 GB) has a recorded licence and sha256, enforced by `tests/test_licenses.py`. The dataset is released under CC BY 4.0; code and the solver wrapper under MIT; Circuitscape.jl 5.17.1 and Omniscape.jl 0.6.2 are MIT. The full attribution block and the Copernicus notices are reproduced in the dataset card and `LICENSE-DATA`.

Table 21: Upstream data sources, their licences and what AmpScape stores (derived rasters only; never the source tiles). WDPA polygons (restrictive licence) and OpenStreetMap (ODbL) are not used. The dataset is released under CC BY 4.0, code and solver wrapper under MIT.

layer	version	licence	derived rasters redistributed
ESA WorldCover land cover, 10 m	2021 v200	CC BY 4.0	class codes
Copernicus DEM GLO-30	2021 release (DLR 2010–2014, Airbus 2014–2018)	Copernicus licence; Art. 4 rights of use, Art. 6 notices reproduced	elevation, slope
GRIP4 global roads	v4, 2018	CC0 by the publisher; CC BY 4.0 in secondary catalogues (attributed as CC BY 4.0)	distance-to-road only
HydroRIVERS	v1.0	CC BY 4.0	distance-to-river only
gHM global Human Modification	v1, 1 km	CC BY 4.0	gHM
RESOLVE Ecoregions 2017	2017	CC BY 4.0	stratification only (biome, realm)
Eurac Alps permeability	2022	CC BY 4.0	published resistance tiles
Hawaiian gallinule resistance	2020	CC0 1.0	published resistance tiles
Northern raccoon Europe maps	2025	CC BY 4.0	published resistance tiles

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   Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results?
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   Justification: Data on the Hugging Face Hub (CC BY 4.0) with Croissant metadata and a `mini` sample; code (MIT) with a README quickstart, the exact evaluation command (Appendix J) and the leaderboard tooling; anonymised at submission.

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